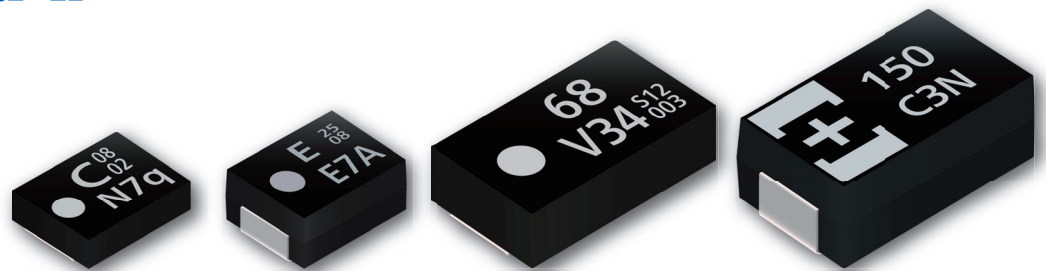


Products Catalog

**Conductive Polymer Tantalum
Solid Capacitors**
POSCAP



**IN Your
Future**



Conductive Polymer Tantalum Solid Capacitors INDEX

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*1: Some part numbers are not a recommended product.

TH series : Not recommended for new design

Safety and Legal Matters to Be Observed

Product specifications and applications

- Please be advised that this product and product specifications are subject to change without notice for improvement purposes. Therefore, please request and confirm the latest delivery specifications that explain the specifications in detail before the final design, or purchase or use of the product, regardless of the application. In addition, do not use this product in any way that deviates from the contents of the company's delivery specifications.
- Unless otherwise specified in this catalog or the product specifications, this product is intended for use in general electronic equipment (AV products, home appliances, commercial equipment, office equipment, information and communication equipment, etc.).
When this product is used for the following special cases, the specification document suited to each application shall be signed/sealed (with Panasonic Industry and the user) in advance..These include applications requiring special quality and reliability, wherein their failures or malfunctions may directly threaten human life or cause harm to the human body (e.g.: space/aircraft equipment, transportation/traffic equipment, combustion equipment, medical equipment, disaster prevention/crime prevention equipment, safety equipment, etc.).

Safety design and product evaluation

- Please ensure safety through protection circuits, redundant circuits, etc., in the customer's system design so that a defect in our company's product will not endanger human life or cause other serious damage.
- This catalog shows the quality and performance of individual parts. The durability of parts varies depending on the usage environment and conditions. Therefore, please ensure to evaluate and confirm the state of each part after it has been mounted in your product in the actual operating environment before use.
If you have any doubts about the safety of this product, then please notify us immediately, and be sure to conduct a technical review including the above protection circuits and redundant circuits at your company.

Laws / Regulations / Intellectual property

- The transportation of dangerous goods as designated by UN numbers, UN classifications, etc., does not apply to this product. In addition, when exporting products, product specifications, and technical information described in this catalog, please comply with the laws and regulations of the countries to which the products are exported, especially those concerning security export control.
- Each model of this product complies with the RoHS Directive (Restriction of the use of hazardous substances in electrical and electronic equipment) (2011/65/EU and (EU) 2015/863). The date of compliance with the RoHS Directive and REACH Regulation varies depending on the product model.
Further, if you are using product models in stock and are not sure whether or not they comply with the RoHS Directive or REACH Regulation, please contact us by selecting "Sales Inquiry" from the inquiry form.
- During the manufacturing process of this product and any of its components and materials to be used, Panasonic Industry does not intentionally use ozone-depleting substances stipulated in the Montreal Protocol and specific bromine-based flame retardants such as PBBs (Poly-Brominated Biphenyls) / PBDEs (Poly-Brominated Diphenyl Ethers). In addition, the materials used in this product are all listed as existing chemical substances based on the Act on the Regulation of Manufacture and Evaluation of Chemical Substances.
- With regard to the disposal of this product, please confirm the disposal method in each country and region where it is incorporated into your company's product and used.
- The technical information contained in this catalog is intended to show only typical operation and application circuit examples of this product. This catalog does not guarantee that such information does not infringe upon the intellectual property rights of Panasonic Industry or any third party, nor imply that the license of such rights has been granted.
- Design, materials, or process related to technical owned by Panasonic Industry are subject to change without notice.

Panasonic Industry will assume no liability whatsoever if the use of our company's products deviates from the contents of this catalog or does not comply with the precautions. Please be advised of these restrictions.

Matters to Be Observed When Using This Product

(Conductive Polymer Tantalum Solid Capacitors / POSCAP)

Use environments and cleaning conditions

- This product (capacitor) is intended for standard general-purpose use in electronic equipment, and is not designed for use in the specific environments described below. Using the product in such specific environments or service conditions, therefore, may affect the performance of the product.

Please check with us about the performance and reliability of the product first before using the product.

- (1) Used in liquid, such as water, oil, chemicals, and organic solvents.
- (2) Used in a place exposed to direct sunlight, an outdoor place with no shielding, or a dusty place.
- (3) Used in a wet place (dew concentration on a resistor, water leakage, etc.), a place exposed to sea breeze, or a place filled with a corrosive gas, such as Cl_2 , H_2S , NH_3 , SO_2 , or NO_x .
- (4) Used in an environment where static electricity and electromagnetic waves are strong.
- (5) The product is located close to a heating component or a flammable material, such as a vinyl cable, is placed near the product.
- (6) The product is used sealed with a resin, etc.
- (7) Solder flux of the soldered product is cleansed with a solvent, water, and a water-soluble cleaner.
(Be careful with water soluble solder flux.)
- (8) Used in an environment where an acidic or alkali atmosphere is present.
- (9) Used in an environment where excessive vibration or impact is applied to the product.
- (10) Used under a low atmospheric pressure condition or depressurized condition.

- When the capacitor is used in a circuit where an impact voltage is applied or a high voltage is applied in a short period (transient phenomenon) or a high pulse voltage is applied, make sure to use the capacitor at a voltage equal to or lower than its rated voltage.

Response to anomalies and handling conditions

- A short mode is a major failure mode in a capacitor. A short mode is caused by thermal stress created by soldering or a high service temperature, electric stress, mechanical stress, etc. When the capacitor has shorted, take the following steps to ensure your safety.
 - (1) When you see smoke coming out of the shorted product, turn off the main power supply to stop using the capacitor.
Do not place your face or hand near to the smoldering capacitor.
 - (2) The time a shorted capacitor takes to generate smoke ranges from a few seconds to a few minutes, depending on service conditions. If you incorporate a protective circuit, design the circuit to activate in the time preceding the smoke generation.
 - (3) In case smoke gets into your eyes or comes into your mouth, wash the eyes with water or rinse out the mouth immediately.
 - (4) When a current value after the short of the product is extremely large, the shorted capacitor may spark out, which, in the worst scenario, may result in ignition. Ensure the safety of the circuit by, for example, giving it a redundant circuit structure or providing it with a protective circuit.

Reliability and product life

The failure rate of the capacitor is specified based on $0.5\%/1000 \text{ h}$ (reliability level 60%), a failure rate conforming to JIS C 5003 (failure rate level). This indicates that the possibility of occurrence of a failure is by no means zero. One of the failure modes is a wear out failure. This happens when the period of guaranteeing the durability and high-temperature/high-humidity resistance of the capacitor is over, changes in the electrical characteristics of the capacitor (product) get larger and its electrolyte gradually deteriorates into an insulating material to create an open mode. Another failure mode is a random failure in which a short mode results mainly because of thermal, electrical, or mechanical stress, etc.

* The failure rate of a small POSCAP of a B2 size or smaller is 1.0%.

Circuit design and circuit board design

- Do not use this capacitor in a high-impedance voltage holding circuit, a coupling circuit, a time constant circuit, or a circuit widely affected by leak current.
- The capacitor is incorporated and used in a circuit where the capacitor operates within a rating range specified in the specifications. Set a service temperature within a specified category temperature range. Do not let a current larger than the allowable ripple current flow in the circuit. Reduce a ripple current to the extent at which the surface temperature of the capacitor's top does not exceed the rated temperature.
(For information about TQC series, please contact us separately.)
- Electrical characteristics values listed in a characteristics table, such as a capacitance and an ESR, are values specified at shipment of the capacitor. These values may change when departing from the specified values in the table under certain electrical or mechanical performance condition. Be careful in choosing a capacitor with the intended electrical characteristics in your design work. Besides, temperature/frequency fluctuations can cause the capacitor's electrical characteristics to change. Confirm such changes in the electrical characteristics and then proceed with your circuit design.
- A leak current may increase even when soldering conditions are within a specified range. The leak current may increase also in a high-temperature non-load test or humidity-resistance non-load test, in which no voltage is applied, a temperature cycle test, etc. In such cases, applying a voltage at a temperature equal to or lower than the highest service temperature of the capacitor reduces the leak current gradually.
- A flow of excessively large rush current created by rapid charge/discharge may result in short circuit or an increase in leak current. When a rush current value exceeds 20 A*, apply a protective circuit to the product.
Note that at leak current measurement, a protective resistance of about 1 kΩ is put in the circuit before the capacitance is charged or discharges.

Mounting and storage conditions

- Set soldering conditions within a specified range. Stricter soldering conditions outside the specified range will cause the deterioration of the electrical characteristics and service life characteristics.
- The capacitor must be kept in storage in an environment that avoids dropping soldering performance or caused trouble with soldering because of moisture absorption by an exterior resin. Store the capacitor, which is put together with a reel in an airtight moisture-proof bag, in a place where a normal temperature and humidity (15 °C to 35 °C and 45%RH to 75%RH) are maintained and direct sunlight is blocked. The storage period is 18 months or less after shipment from the factory.
- Unseal the bag to take out the capacitor right before mounting it on a circuit. Once you take out the capacitor, make sure to use it up. Storage periods for capacitors taken out of bags are as follows.
 - * These products do not conform to the entire requirements defined in JEDEC J-STD-020 and J-STD-033.
 - (1) Level 2a : four weeks at a temperature and a humidity equal to or lower than 30 °C and 60 %RH
 - (2) Level 3 : 168 hours at a temperature and a humidity equal to or lower than 30 °C and 60 %RH
 - (3) Level 5 : 48 hours at a temperature and a humidity equal to or lower than 30 °C and 60 %RH

Reference information

Intellectual property

Panasonic Group provides customers with safe products and services. We are also making great efforts to protect our intellectual property rights for Panasonic Group products. Typical patents related to this product are as follows.

[U.S. patent]

USP Nos. 8081421, 8149569, 8456804, and 8559166

Line up

Series	Features	Small size/Low profile	Large capacitance	Low ESR	High reliability	High voltage	Guaranteed at 125°C	Category temp. range (°C)	Rated voltage (V)	ESR (mΩ)	Capacitance (μF)	Size code	Size (mm)		
													L	W	H
TPS	Small size/Low profile Large capacitance Face down terminal	●	●					-55 to 105	2.5 to 6.3	30 to 35	150 to 330	B1S	3.5	2.8	1.1
TPSF	Low ESR/Small size Large capacitance Face down terminal	●	●	●				-55 to 105	2.0 to 2.5	6 to 9	270	B2S	3.5	2.8	1.9
TPE ^{*1}	Low ESR				●			-55 to 105	2.0 to 10	9 to 70	47 to 470	B2	3.5	2.8	1.9
								-55 to 105	2.5 to 10	7 to 45	68 to 470	D2E	7.3	4.3	1.8
								-55 to 105	2.5 to 10	9 to 40	150 to 680	D3L	7.3	4.3	2.8
								-55 to 105	2.5 to 10	10 to 40	330 to 1500	D4	7.3	4.3	3.8
TPF ^{*1}	Low ESR Large capacitance		●	●				-55 to 105	2.5 to 10	5 to 25	150 to 680	D3L	7.3	4.3	2.8
								-55 to 105	2.5 to 6.3	5 to 35	470 to 1000	D4	7.3	4.3	3.8
TQT	Super high voltage Low profile Large capacitance	●	●			●		-55 to 105	35	100	56	D15S	7.3	4.3	1.4
TQS	High voltage	●				●		-55 to 105	35	100	47	D15S	7.3	4.3	1.4
								-55 to 105	35	100	68	D2S	7.3	4.3	1.9
TQC ^{*1}	High voltage					●		-55 to 105	16 to 35	90 to 200	10 to 47	B2	3.5	2.8	1.9
								-55 to 105	16 to 25	55 to 70	22 to 47	D15	7.3	4.3	1.4
								-55 to 105	16 to 35	50 to 180	10 to 150	D2	7.3	4.3	1.9
								-55 to 105	16 to 25	50 to 70	68 to 220	D3L	7.3	4.3	2.8
TA	High reliability				●			-55 to 105	4.0 to 10	70	47 to 100	B2	3.5	2.8	1.9
								-55 to 105	2.5 to 6.3	9 to 25	150 to 470	D2E	7.3	4.3	1.8
								-55 to 105	2.5 to 10	15 to 25	150 to 680	D3L	7.3	4.3	2.8
TV	High reliability Guaranteed at 125 °C				●	●		-55 to 125	6.3 to 10	25	68 to 150	D2E	7.3	4.3	1.8
								-55 to 125	10	25	150	D3L	7.3	4.3	2.8
TPB ^{*1}	Standard							-55 to 105	4.0 to 10	70	33 to 68	B2	3.5	2.8	1.9
								-55 to 105	4.0 to 10	40 to 55	150 to 330	D3L	7.3	4.3	2.8
								-55 to 105	6.3 to 10	35 to 40	220 to 470	D4	7.3	4.3	3.8

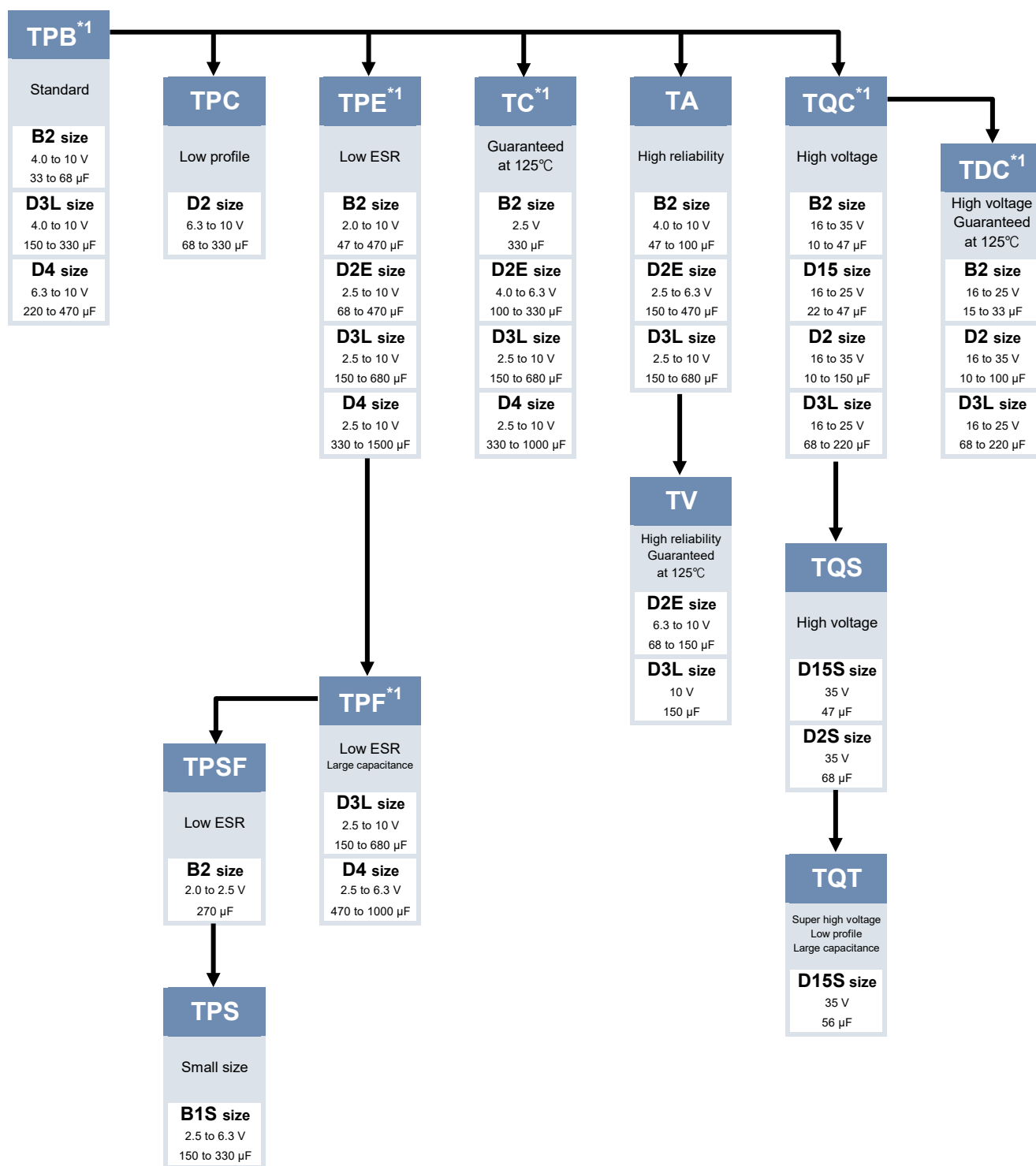
^{*1} : Some part numbers are not a recommended product

Line up

Series	Features	Small size/Low profile	Large capacitance	Low ESR	High reliability	High voltage	Guaranteed at 125°C	Category temp. range (°C)	Rated voltage (V)	ESR (mΩ)	Capacitance (μF)	Size code	Size (mm)		
													L	W	H
TC ^{*1}	Guaranteed at 125 °C							-55 to 125	2.5	9	330	B2	3.5	2.8	1.9
							●	-55 to 125	4.0 to 6.3	15 to 25	100 to 330	D2E	7.3	4.3	1.8
								-55 to 125	2.5 to 10	5 to 25	150 to 680	D3L	7.3	4.3	2.8
								-55 to 125	2.5 to 10	5 to 25	330 to 1000	D4	7.3	4.3	3.8
TDC ^{*1}	High voltage Guaranteed at 125 °C							-55 to 125	16 to 25	90 to 100	15 to 33	B2	3.5	2.8	1.9
						●	●	-55 to 125	16 to 35	50 to 180	10 to 100	D2	7.3	4.3	1.9
								-55 to 125	16 to 25	50 to 70	68 to 150	D3L	7.3	4.3	2.8
TPC ^{*1}	Low profile	●						-55 to 105	10	45 to 100	68 to 100	D2	7.3	4.3	1.9

*1 : Some part numbers are not a recommended product

Series flow chart



:1: Some part numbers are not a recommended product

Voltage - Capacitance table (Vol. : 2.0 to 8.0 V / Cap. : 3.9 to 68 μ F)

Series [Size]
(ESR m Ω)

V μ F	3.9	4.7	5.6	6.8	8.2	10	15	22	33	47	56
2.0											
2.5											
4.0											
6.3									TPB [B2] (70)	TA [B2] (70)	
8.0											

Size list LxWxH (mm)

B1	3.5x2.8x1.1	B2	3.5x2.8x1.9	D15	7.3x4.3x1.4	D2E	7.3x4.3x1.8	D3L	7.3x4.3x2.8
B1S	3.5x2.8x1.1	B2S	3.5x2.8x1.9	D15S	7.3x4.3x1.4	D2	7.3x4.3x1.9	D4	7.3x4.3x3.8
						D2S	7.3x4.3x1.9		

Voltage - Capacitance table (Vol. : 2.0 to 8.0 V / Cap. : 100 to 1500 µF)

Series [Size]
(ESR mΩ)

V	µF	68	100	120	150	220	270	330	470	680	1000	1500
2.0						TPSF [B2S] (9,6)			TPE [B2] (15,11)			
2.5						TPS [B1S] (30)	TPSF [B2S] (6)	TPS [B1S] (30)	TPE [D2E] (18,15,12,9,7)	TPE [D3L] (40,15,12)	TPE [D4] (15)	TPE [D4] (15,12)
						TPE [B2] (35,30,25,21,15)		TPE [B2] (35,18,9)	TPF [D3L] (10,7,6)	TPF [D3L] (10,9,7,6)	TPF [D4] (6,5)	
						TPE [D2E] (18,15,9)		TPE [D2E] (18,15,12,9,7)	TA [D2E] (25,15)	TPF [D4] (25)	TC [D4] (15,6,5)	
						TA [D2E] (25,15,9)		TA [D2E] (25,18,15)		TA [D3L] (25,15)		
								TC [B2] (9)		TC [D3L] (15,12,10,7,6)		
4.0		TPB [B2] (70)	TPE [B2] (70,40,35)		TPE [B2] (35)	TPS [B1S] (35,30)		TPE [D2E] (40,25,18)	TPE [D3L] (40,25,18,15,12)	TPF [D4] (35,15,10)		
			TA [B2] (70)		TPE [D2E] (18)	TPE [B2] (35)		TPF [D3L] (15,12,9)	TPF [D3L] (10)	TC [D4] (10)		
					TC [D2E] (18)	TPE [D2E] (45,25,18,15)		TC [D2E] (25,18)	TA [D3L] (25,18)			
						TA [D2E] (25,18)		TC [D3L] (12)	TC [D3L] (25,18,15,12,10)			
						TC [D2E] (25,18,15)						
6.3		TA [B2] (70)	TPE [B2] (70,45,40,35,25)	TPE [B2] (35)	TPS [B1S] (35,30)	TPE [B2] (35,25)		TPE [D2E] (25)	TPE [D4] (40,35,25,18)	TPE [D4] (25,18)		
		TPB [B2] (70)	TPE [D2E] (45,25,18)		TPE [B2] (35,25)	TPE [D2E] (40,25,18)		TPE [D3L] (25,18,15)	TPF [D4] (15,10)	TC [D4] (25,18)		
					TPE [D2E] (25,18,15)	TPE [D3L] (18)		TPF [D3L] (10,9)	TPB [D4] (35)			
					TPF [D3L] (25)	TPF [D3L] (25,15,12)		TA [D3L] (25)	TC [D4] (25,18,10)			
					TA [D2E] (25)	TA [D2E] (25,18)		TPB [D3L] (45,40)				
					TV [D2E] (25)	TC [D2E] (25,18)		TPB [D4] (40)				
					TC [D2E] (25,18,15)	TC [D3L] (12)		TC [D3L] (25,18,15,9)				
8.0			TPE [B2] (35)									

Size list LxWxH (mm)

B1	3.5x2.8x1.1	B2	3.5x2.8x1.9	D15	7.3x4.3x1.4	D2E	7.3x4.3x1.8	D3L	7.3x4.3x2.8
B1S	3.5x2.8x1.1	B2S	3.5x2.8x1.9	D15S	7.3x4.3x1.4	D2	7.3x4.3x1.9	D4	7.3x4.3x3.8
						D2S	7.3x4.3x1.9		

Voltage - Capacitance table (Vol. : 10 to 63 V / Cap. : 3.9 to 68 μF)Series [Size]
(ESR m Ω)

V	μF	3.9	4.7	5.6	6.8	8.2	10	15	22	33	47	56
10										TPB [B2] (70)	TPE [B2] (35)	
											TA [B2] (70)	
											TPB [B2] (70)	
16										TQC [B2] (90)	TQC [B2] (90)	
										TDC [B2] (90)	TQC [D15] (55)	
											TQC [D2] (55,70)	
											TDC [D2] (55,70)	
20									TQC [B2] (90)	TQC [D2] (60)	TQC [D15] (55)	
									TQC [D2] (90)	TQC [B2] (100)	TQC [D2] (55)	
									TDC [B2] (90)	TDC [D2] (60)	TDC [D2] (55)	
25								TQC [B2] (100)	TQC [B2] (100)	TQC [D2] (60)		
								TQC [D2] (90)	TQC [D2] (60)	TDC [D2] (60)		
								TDC [B2] (100)	TDC [D2] (60)			
								TDC [D2] (90)				
35							TQC [B2] (200,150)	TQC [D2] (150)	TQC [D2] (150)		TQS [D15S] (100)	TQT [D15S] (100)
							TQC [D2] (120,180)	TDC [D2] (150)	TDC [D2] (150)			
							TDC [D2] (180)					

Size list LxWxH (mm)

B1	3.5x2.8x1.1	B2	3.5x2.8x1.9	D15	7.3x4.3x1.4	D2E	7.3x4.3x1.8	D3L	7.3x4.3x2.8
B1S	3.5x2.8x1.1	B2S	3.5x2.8x1.9	D15S	7.3x4.3x1.4	D2	7.3x4.3x1.9	D4	7.3x4.3x3.8
						D2S	7.3x4.3x1.9		

Voltage - Capacitance table (Vol. : 10 to 63 V / Cap. : 100 to 1500 μ F)

Series [Size]
(ESR m Ω)

V	μ F	68	100	120	150	220	270	330	470	680	1000	1500
10		TV [D2E] (25)	TPC [D2] (55,50,45)		TPE [D3L] (25)	TPE [D3L] (25,18)		TPE [D4] (25)				
		TPC [D2] (100,60,45)			TPF [D3L] (15)	TA [D3L] (25)		TPB [D4] (40)				
					TA [D3L] (25)	TPB [D3L] (40)		TC [D4] (25)				
					TV [D3L] (25)	TPB [D4] (40)						
					TPB [D3L] (55,40)	TC [D3L] (25,18)						
					TC [D3L] (15)							
16		TQC [D2] (50)	TQC [D2] (50)		TQC [D2] (50)	TQC [D3L] (50)						
		TDC [D2] (50)	TDC [D2] (50)		TQC [D3L] (50)							
					TDC [D3L] (50)							
20			TQC [D2] (100)									
			TQC [D3L] (55)									
			TDC [D3L] (55)									
25		TQC [D3L] (70)	TQC [D3L] (60)									
		TDC [D3L] (70)										
35		TQS [D2S] (100)										

Size list LxWxH (mm)

B1	3.5x2.8x1.1	B2	3.5x2.8x1.9	D15	7.3x4.3x1.4	D2E	7.3x4.3x1.8	D3L	7.3x4.3x2.8
B1S	3.5x2.8x1.1	B2S	3.5x2.8x1.9	D15S	7.3x4.3x1.4	D2	7.3x4.3x1.9	D4	7.3x4.3x3.8
						D2S	7.3x4.3x1.9		

Explanation of part numbers

◆ Part number system

2R5		TPB		330		M		L	
Rated voltage 1 to 3 figures		Series name 3 to 4 figures		Rated cap. 2 to 4 figures		Cap. tolerance 1 figure		Special code 0 to 4 figures	
R.voltage (V)	Code	Series	Code	R. cap. (μF)	Code	Cap. tolerance	Code		
2.0	2	TPB	TPB	3.9	3R9	±20%	M		
2.5	2R5 or E	TPC	TPC	4.7	4R7				
4.0	4	TPE	TPE	5.6	5R6				
6.3	6	TPF	TPF	6.8	6R8				
8.0	8	TPS	TPS	8.2	8R2				
10.0	10	TPSF	TPSF	10	10				
12.5	12	TAB	TAB	15	15				
16.0	16 or 1C	TAE	TAE	22	22				
20.0	20	TQC	TQC	33	33				
25.0	25	TQS	TQS	47	47				
35.0	35	TVE	TVE	56	56				
50.0	50	TCE	TCE	68	68				
63.0	63	TCF	TCF	82	82				
		TDC	TDC	100	100				
		TQT	TQT	120	120				
				150	150				
				220	220				
				270	270				
				330	330				
				470	470				
				680	680				
				1000	1000				
				1500	1500				

Standard		Code
TPE series		
B2	ESR 35 mΩ max.	ZB
	ESR 25 mΩ max.	PB
	ESR 21 mΩ max.	LB
	ESR 15 mΩ max.	FB
	ESR 15 mΩ / 300 kHz max.	FGB
	ESR 35 mΩ max. 85°C	AZB
	ESR 25 mΩ max. 85°C	APB
	ESR 15 mΩ max. 85°C	AFB
	ESR 13 mΩ / 300 kHz max. 85°C	ADGB
D2E	ESR 11 mΩ / 300 kHz max. 85°C	AJGB
	ESR 25 mΩ max. 85°C	AP
D3L	ESR 25 mΩ max.	L
	ESR 18 mΩ max.	IL
	ESR 15 mΩ max.	FL
	ESR 12 mΩ max.	CL
	ESR 10 mΩ max.	AL
	ESR 25 mΩ max. 85°C	AL
ESR 9 mΩ / 500 kHz max. 85°C		A9EL
TPB series		
D3L		L
TPC series		
85 °C		A
B1		B
TQS / TQT series		
D15S	ESR 100 mΩ / Height 1.5 mm max.	EU
D2S	ESR 100 mΩ / Height 2.0 mm max.	E2

Standard		Code
TPF series		
D3L	ESR 9 mΩ max.	9L
	ESR 7 mΩ max.	7L
	ESR 6 mΩ max.	6L
	ESR 5 mΩ max.	5L
D4	ESR 10 mΩ max.	AH
	ESR 6 mΩ max.	6H
	ESR 5 mΩ max.	5H
TQC / TDC series		
Capacitance enlarged type		YF
Capacitance enlarged type (B2 size)		YFB
Capacitance enlarged type (D15 size)		YFT
Capacitance enlarged type (D2 size)		YFD/D2
Capacitance enlarged type (D3L size)		D3
All series		
ESR 55 mΩ max.		G
ESR 45 mΩ max.		V
ESR 40 mΩ max.		W
ESR 35 mΩ max.		Z
ESR 18 mΩ max.		I
ESR 15 mΩ max.		F
ESR 12 mΩ max.		C
ESR 9 mΩ max.		9
ESR 7 mΩ max.		7
ESR 6 mΩ max.		6
ESR 5 mΩ max.		5
ESR 35 mΩ / 300 kHz max.		ZG
ESR 30 mΩ / 300 kHz max.		UG
ESR 9 mΩ / 300 kHz max.		9G
ESR 6 mΩ / 500 kHz max.		6E
ESR 4 mΩ / 500 kHz max.		4E

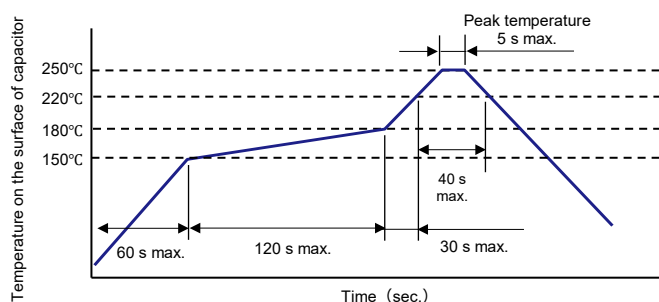
TPE/TPF/TPC/TQC series some part numbers : Not recommended for new design

Mounting specifications

◆ Recommendable reflow soldering

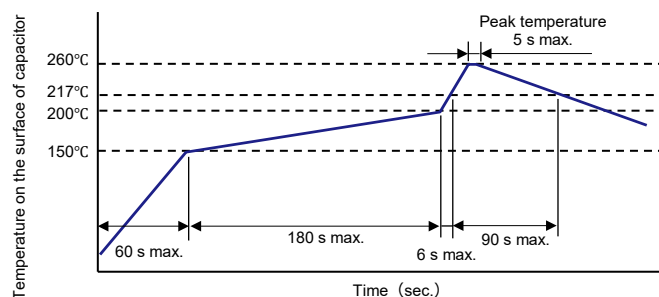
■ Applicable series : TPS, TPSF, TPE, TPF, TA
TV, TPB, TC, TPC

Peak temperature 250 °C lead free reflow
soldering profile
The cycles of reflow soldering : Twice (max.)



■ Applicable series : TPS, TPSF, TPE, TPF, TA
TV, TPB, TC, TPC

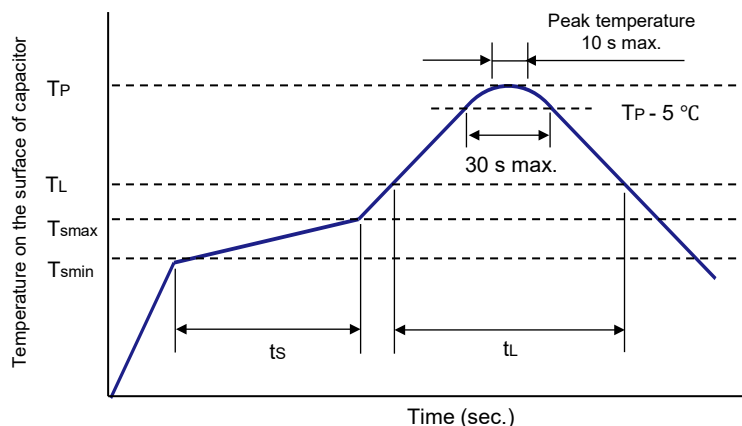
Peak temperature 260 °C lead free reflow
soldering profile
The cycles of reflow soldering : Twice (max.)



Note) This reflow is limited at moisture sensitive level.
Please contact us separately concerning about detail.

■ Applicable series : TQC, TDC, TQS, TQT

* Refer to the characteristics list of each
series for compatible part numbers.



Reflow soldering		Condition A	Condition B	Condition C	Condition D
		IPC/J-STD-020 reflow profile		Peak temperature 250 °C lead free reflow profile	
Peak temperature (T _P)		260 °C	250 °C	250 °C	
Time within -5 °C of peak temperature		30 seconds max.		—	
Soldering time (t _L) above soldering temperature (T _L)	T _L ≥ 217 °C	150 seconds max.		90 seconds max.	60 seconds max.
	T _L ≥ 200 °C	—			70 seconds max.
Preheat	Temperature max. (T _{smax})	200 °C		180 °C	
	Temperature min. (T _{smin})	150 °C			
	Time (t _s) from (T _{smin}) to (T _{smax})	120 seconds max.			
The cycles of reflow soldering		3 times (max.)		Twice (max.)	

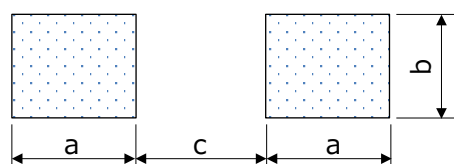
● Soldering with a soldering iron

Tip of a soldering iron : 350 °C max (TQC / TDC / TQS / TQT series : 400 °C max.)

Working time : 3 sec. max. (TQC / TQS / TQT series : 5 sec. max.)

(Do not let the tip of soldering iron touch the POSCAP itself. Do not subject the POSCAP itself to excessive stress when soldering)

Land pattern

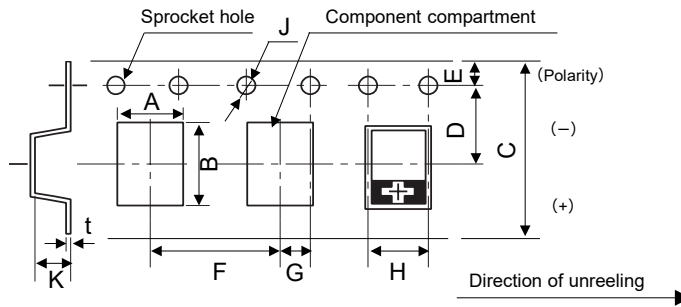


Size code	a	b	c
B1, B1S, B2, B2S	1.6	2.7	1.4
D15, D2E, D2, D3L, D4	2.4	2.9	3.7
D15S	1.4 / 1.7 ^{*1}	2.6	4.6
D2S	1.4	2.6	5.1

*1 : +side / -side

Packing specifications

◇ Dimension of carrier tape



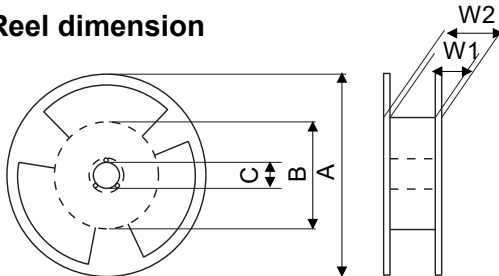
Unit : mm

Size code	A±0.2	B±0.2	C±0.3	D±0.1	E±0.1	F±0.1	G±0.1	H±0.1	J $\begin{smallmatrix} +0.1 \\ -0 \end{smallmatrix}$	K±0.1	t±0.1
B1	3.2	3.8	8.0	3.5	1.75	4.0	2.0	4.0	ø1.5	1.4	0.25
B1S	3.25	3.9	8.0	3.5	1.75	4.0	2.0	4.0	ø1.5	1.7	0.25
B2	3.3	3.8	8.0	3.5	1.75	4.0	2.0	4.0	ø1.5	2.1	0.25
B2S	3.25	4.0	8.0	3.5	1.75	4.0	2.0	4.0	ø1.5	2.1	0.25
D12	4.5	7.5	12.0	5.5	1.75	8.0	2.0	4.0	ø1.5	1.7	0.3
D15	4.5	7.5	12.0	5.5	1.75	8.0	2.0	4.0	ø1.5	2.4	0.3
D15S	4.7	7.8	12.0	5.5	1.75	8.0	2.0	4.0	ø1.5	1.7	0.3
D2E	4.5	7.5	12.0	5.5	1.75	8.0	2.0	4.0	ø1.5	2.4	0.3
D2	4.5	7.5	12.0	5.5	1.75	8.0	2.0	4.0	ø1.5	2.4	0.3
D2S	4.5	7.8	12.1	5.5	1.75	8.0	2.0	4.0	ø1.5	2.4	0.3
D3L	4.5	7.7	12.0	5.5	1.75	8.0	2.0	4.0	ø1.5	3.2	0.3
D4	4.5	7.7	12.0	5.5	1.75	8.0	2.0	4.0	ø1.5	4.2	0.3

- Dimension A and B are the measure of compartment's inside bottom.
- The (+) Polarity of the chip is placed on right side towards the unreeling direction.
- Dimension of the topcover tape.
Thickness of cover tape: 62±10 µm, Width of cover tape : 9.5±0.2 mm 5.5±0.2 mm (ø180)

D12 size : Not recommended for new design

◇ Reel dimension



Unit : mm

A	B	C	W1	W2
ø330±2	ø80±2	ø13.0±0.2	13.5±0.5	17.5±1.0
ø180 $\begin{smallmatrix} 0 \\ -3 \end{smallmatrix}$	ø60±2	ø13.0±0.2	9.0±0.5	11.4±1.0

◇ Minimum packing quantity and weight

〈ø180〉

Size code	Qty. (pcs./Reel)	Typical weight (g)
B1	3000	200
B1S	2500	200
B2, B2S	2000	200

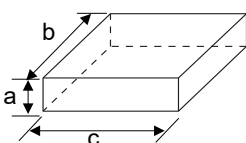
〈ø330〉

Size code	Qty. (pcs./Reel)	Typical weight (g)
D12	4500	1200
D15	3000	1000
D15S	4000	1000
D2E, D2, D2S	3000	1000
D3L	2500	1100
D4	2000	1200

※ Small order quantity (500 pcs/reel) is available with TPE, TPF and TQC series.
Please contact our sales representative if you prefer it.

D12 size : Not recommended for new design

◇ Dimension of packing case



Unit : mm

Reel size	ø180	ø330
a	90	120
b	240	360
c	240	360

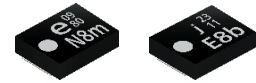
◇ Units per packing case

Size code	Pieces/case	Size code	Pieces/case
B1	15000	D12	22500
B1S	12500	D15	15000
B2, B2S	10000	D15S	20000
		D2E, D2, D2S	15000
		D3L	12500
		D4	10000

D12 size

: Not recommended for new design

Conductive Polymer Tantalum Solid Capacitors Surface Mount Type TPS series



Features

- 105 °C 2000 h
- Small size, Low profile (L 3.5 × W 2.8 × H 1.2 mm max.)
- RoHS compliance, Halogen free

Specifications

Size code	B1S	
Category temp. range	-55 °C to +105 °C	
Rated voltage range	2.5 V to 6.3 V	
Category voltage range	2.5 V to 6.3 V	
Rated capacitance range	150 μF to 330 μF	
Capacitance tolerance	±20 % (120 Hz / +20 °C)	
Leakage current	Please see the attached characteristics list	
Dissipation factor(tan δ)	Please see the attached characteristics list	
Surge voltage (V)	Rated voltage × 1.15	
Endurance	+105 °C 2000 h rated voltage applied	
	Capacitance change	Within ±20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	Within the initial limit
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage	
	Capacitance change	Within +40 %, -20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	≤ 3 times of the initial limit

Marking

R.voltage code	Unit : V
e	2.5
g	4.0
j	6.3
R.capacitance code	Unit : μF
E8	150
J8	220
N8	330

Dimensions (not to scale)

Unit : mm	
Size code	L±0.2 W±0.2 H±0.1 S±0.3 W1±0.1
B1S	3.5 2.8 1.1 0.8 2.2

* Externals of figure are the reference.

Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	
					L	W	H		Ripple ^{*1} current (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (μA)	Part number	Min. packaging q'ty (pcs)	Reflow temp ≤260°C	Reflow temp ≤250°C
2.5	105	2.5	105	220	3.5	2.8	1.1	B1S	1400	30	0.10	55.0	ETPS220MUD	2500	3	3
	105	2.5	105	330	3.5	2.8	1.1		1400	30	0.10	82.5	ETPS330MUD	2500		
4.0	105	4.0	105	220	3.5	2.8	1.1		1400	30	0.10	88.0	4TPS220MUD	2500		
	105	6.3	105	150	3.5	2.8	1.1		1400	30	0.10	94.5	6TPS150MUD	2500		
6.3	105	6.3	105		3.5	2.8	1.1		1250	35	0.10	94.5	6TPS150MZD	2500		

*1: Ripple current (100 kHz / +45 °C)

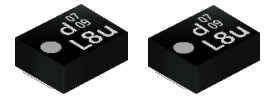
*2: ESR (100 kHz / +20 °C)

*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".

Conductive Polymer Tantalum Solid Capacitors Surface Mount Type TPSF series



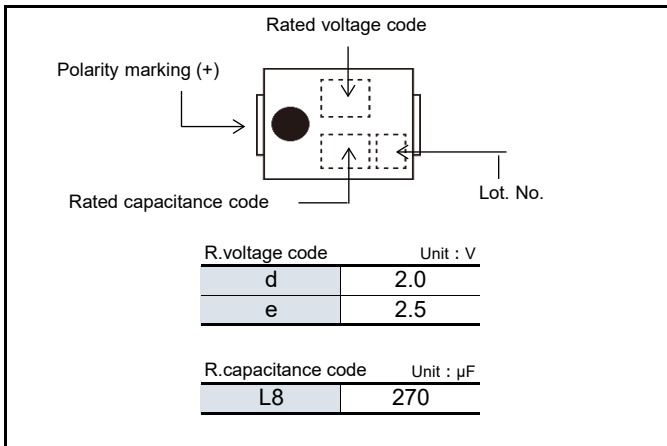
Features

- Super low ESR (6 mΩ max.)
- Super low ESL (0.7 nH max.)
- Face down terminal type
- RoHS compliance, Halogen free

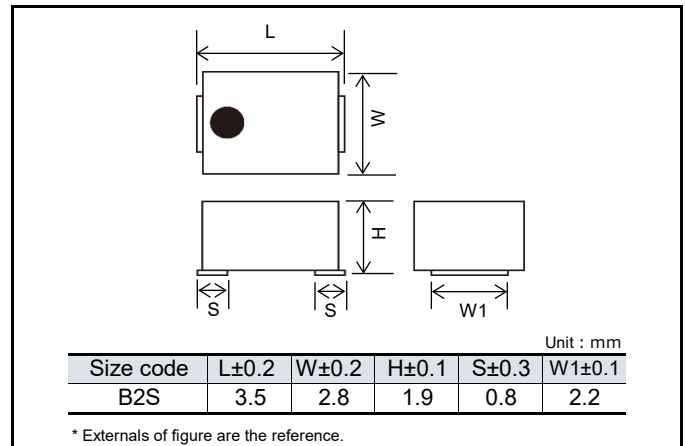
Specifications

Size code	B2S	
Category temp. range	-55 °C to +105 °C	
Rated voltage range	2.0 V to 2.5 V	
Category voltage range	2.0 V to 2.5 V	
Rated capacitance range	270 μF	
Capacitance tolerance	±20 % (120 Hz / +20 °C)	
Leakage current	Please see the attached characteristics list	
Dissipation factor(tan δ)	Please see the attached characteristics list	
Surge voltage (V)	Rated voltage × 1.15	
Endurance	+105 °C 1000 h rated voltage applied	
	Capacitance change	Within ±20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	Within the initial limit
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage	
	Capacitance change	Within +40 %, -20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	≤ 3 times of the initial limit

Marking



Dimensions (not to scale)



Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	
					L	W	H		Ripple current ^{*1} (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (μA)	Part number	Min. packaging q'ty (pcs)	Reflow temp ≤260°C	Reflow temp ≤250°C
2.0	105	2.0	105	270	3.5	2.8	1.9	B2S	3200	6/500 kHz	0.08	108.0	2TPSF270M6E	2000	5	5
	105	2.0	105		3.5	2.8	1.9		2400	9/300 kHz	0.08	108.0	2TPSF270M9G	2000		
2.5	105	2.5	105		3.5	2.8	1.9		3200	6/500 kHz	0.08	135.0	ETPSF270M6E	2000	3	3

*1: Ripple current (100 kHz / +45 °C)

*2: ESR (100 kHz / +20 °C)

*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".



Conductive Polymer Tantalum

Solid Capacitors

Surface Mount Type

TPE series

B size

Features

- Small size (L 3.5 × W 2.8 × H 1.9 mm)
- Low ESR (15 mΩ max.)
- RoHS compliance, Halogen free

Specifications

Size code	B2	
Category temp. range	-55 °C to +105 °C	
Rated voltage range	2.0 V to 10 V	
Category voltage range	1.8 V to 8 V	
Rated capacitance range	47 μF to 470 μF	
Capacitance tolerance	±20 % (120 Hz / +20 °C)	
Leakage current	Please see the attached characteristics list	
Dissipation factor(tan δ)	Please see the attached characteristics list	
Surge voltage (V)	Rated voltage × 1.15	
Endurance	+105 °C 1000 h rated voltage applied	
	Rated temp, +85 °C Products : +85 °C 1000 h, rated voltage applied	
	Capacitance change	Within ±20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
Damp heat (Steady State)	Leakage current	Within the initial limit
	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage	
	Capacitance change	Within +50 %, -20 % of the initial value (2R5TPE220MAZB (MAPB, MAFB), 2R5TPE330MAZB, 2TPE470MAJGB (MAFB))
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	≤ 3 times of the initial limit

Marking

Rated capacitance code

Polarity marking (+)

Rated voltage code

Lot No.

R. voltage code		Unit : V	
d	2.0	g	4.0
e	2.5	j	6.3
		k	8.0
		A	10

R. capacitance code		Unit : μF	
S7	47	E8	150
A8	100	J8	220
C8	120	N8	330
		S8	470

Dimensions (not to scale)

Size code	L±0.2	W±0.2	H±0.1	S±0.2	W1±0.1
B2	3.5	2.8	1.9	0.8	2.2

Unit : mm

* Externals of figure are the reference.

Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	
					L	W	H		Ripple ^{*1} current (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (μA)	Part number	Min. packaging q'ty (pcs)	Reflow temp ≤260°C	Reflow temp ≤250°C
2.0	85	1.8	105	470	3.5	2.8	1.9	B2	2300	15	0.10	188.0	2TPE470MAFB	2000	3	3
	85	1.8	105		3.5	2.8	1.9		2300	11/300kHz	0.08	188.0	2TPE470MAJGB	2000		
2.5	85	2.0	105	220	3.5	2.8	1.9		2000	15	0.08	110.0	2R5TPE220MAFB	2000		
	105	2.5	105		3.5	2.8	1.9		1800	15/300kHz	0.08	110.0	2R5TPE220MFGB	2000		
	105	2.5	105		3.5	2.8	1.9		1700	21	0.08	55.0	2R5TPE220MLB	2000		
	85	2.0	105		3.5	2.8	1.9		1600	25	0.08	55.0	2R5TPE220MAPB	2000		
	105	2.5	105		3.5	2.8	1.9		1400	35	0.08	55.0	2R5TPE220MZB	2000		
	85	2.0	105		3.5	2.8	1.9		1400	35	0.08	55.0	2R5TPE220MAZB	2000		
	105	2.5	105		3.5	2.8	1.9		1400	30	0.08	55.0	2R5TPE220MUB	2000		
	85	2.0	105	330	3.5	2.8	1.9		1400	35	0.08	82.5	2R5TPE330MAZB	2000		
	85	2.0	105		3.5	2.8	1.9		3200	9/300kHz	0.08	165.0	ETPE330MA9GB	2000		
	105	2.5	105		3.5	2.8	1.9		3200	9/300kHz	0.08	165.0	ETPE330M9GB	2000		
	105	2.5	105		3.5	2.8	1.9		2700	15	0.08	165.0	ETPE330MFB	2000		
	105	2.5	105		3.5	2.8	1.9		2450	18	0.08	165.0	ETPE330MIB	2000		
	105	2.5	105		3.5	2.8	1.9		1400	35	0.08	40.0	4TPE100MZB	2000		
4.0	105	4.0	105	100	3.5	2.8	1.9		950	70	0.08	40.0	4TPE100MBB	2000		
	105	4.0	105		3.5	2.8	1.9		1300	40	0.08	40.0	4TPE100MWB	2000		
	105	4.0	105		3.5	2.8	1.9		1400	35	0.08	60.0	4TPE150MAZB	2000		
	85	3.2	105	220	3.5	2.8	1.9		1400	35	0.08	88.0	4TPE220MAZB	2000		
	NRFND	105	4.0		3.5	2.8	1.9		1350	35	0.10	88.0	4TPE220MZB	2000		
	NRFND	105	4.0		3.5	2.8	1.9		1150	45	0.10	88.0	4TPE220MVB	2000		
	NRFND	105	4.0		3.5	2.8	1.9		950	70	0.10	88.0	4TPE220MBB	2000		
	105	6.3	105	100	3.5	2.8	1.9		1600	25	0.08	63.0	6TPE100MPB	2000		
	85	5.0	105		3.5	2.8	1.9		1400	35	0.08	63.0	6TPE100MAZB	2000		
	105	6.3	105		3.5	2.8	1.9		1400	35	0.08	63.0	6TPE100MZB	2000		
6.3	105	6.3	105		3.5	2.8	1.9		950	70	0.08	63.0	6TPE100MBB	2000		
	105	6.3	105		3.5	2.8	1.9		1300	40	0.08	63.0	6TPE100MWB	2000		
	105	6.3	105		3.5	2.8	1.9		1200	45	0.08	63.0	6TPE100MVB	2000		
	85	5.0	105	120	3.5	2.8	1.9		1400	35	0.08	75.6	6TPE120MAZB	2000		
	85	5.0	105		3.5	2.8	1.9		1600	25	0.08	94.5	6TPE150MAPB	2000		
	85	5.0	105		3.5	2.8	1.9		1400	35	0.08	94.5	6TPE150MAZB	2000		
	85	5.0	105		3.5	2.8	1.9		1400	35	0.10	138.6	6TPE220MAZB	2000		
	85	5.0	105	150	3.5	2.8	1.9		1600	25	0.10	138.6	6TPE220MAPB	2000		
	NRFND	105	6.3		3.5	2.8	1.9		1150	45	0.10	138.6	6TPE220MVB	2000		
	NRFND	105	6.3		3.5	2.8	1.9		950	70	0.10	138.6	6TPE220MBB	2000		
	NRFND	105	6.3		3.5	2.8	1.9		1350	35	0.10	138.6	6TPE220MZB	2000		
8.0	85	6.3	105	100	3.5	2.8	1.9		1400	35	0.08	80.0	8TPE100MAZB	2000		
10	85	8.0	105	47	3.5	2.8	1.9		1400	35	0.08	47.0	10TPE47MAZB	2000		

*1: Ripple current (100 kHz / +45 °C)

NRFND Not recommended for new design

*2: ESR (100 kHz / +20 °C)

*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".

◆ Small order quantity (500 pcs/reel) is available with TPE series. Please contact our sales representative if you prefer it.



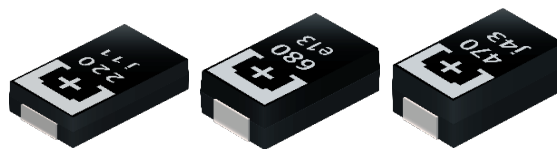
Conductive Polymer Tantalum

Solid Capacitors

Surface Mount Type

TPE series

D size



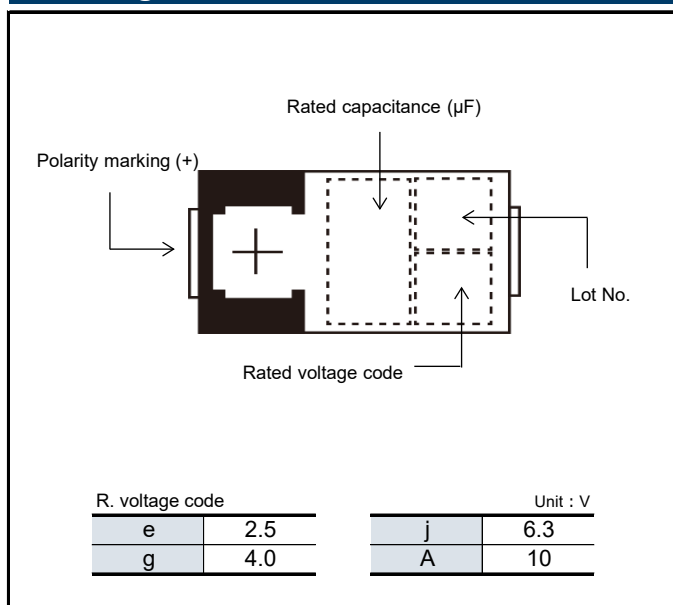
Features

- Low ESR (7 mΩ max.)
- Large capacitance (1500 μF max.)
- RoHS compliance, Halogen free

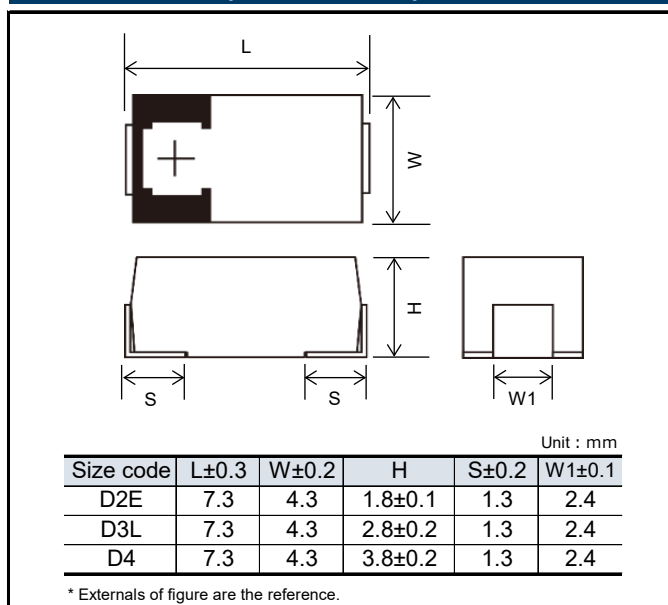
Specifications

Size code	D2E	D3L	D4
Category temp. range	-55 °C to +105 °C		
Rated voltage range	2.5 V to 10 V		
Category voltage range	2.5 V to 10 V		
Rated capacitance range	68 μF to 470 μF	150 μF to 680 μF	330 μF to 1500 μF
Capacitance tolerance	±20 % (120 Hz / +20 °C)		
Leakage current	Please see the attached characteristics list		
Dissipation factor(tan δ)	Please see the attached characteristics list		
Surge voltage (V)	Rated voltage × 1.15		
Endurance	+105 °C 2000 h rated voltage applied Rated temp, +85 °C Products : +85 °C 1000 h, rated voltage applied (6TPE330MAP : +85 °C 2000 h, rated voltage applied)		
	Capacitance change	Within ±20 % of the initial value	
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit	
	Leakage current	Within the initial limit	
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage		
	Capacitance change	Within +50 %, -20 % of the initial value (2R5TPE220M (I, F, 9), 2R5TPE330M (I, F, C, 9, 7), 2R5TPE470M (I, F, C, 9, 7), 2R5TPE1000MF, 2R5TPE1500M (F, C))	
		Within +40 %, -20 % of the initial value (Except for above model)	
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit	
Leakage current	≤ 3 times of the initial limit		

Marking



Dimensions (not to scale)



Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	
					L	W	H		Ripple ^{*1} current (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (μA)	Part number	Min. packaging q'ty (pcs)	Reflow temp ≤260°C	Reflow temp ≤250°C
2.5	105	2.5	105	220	7.3	4.3	1.8	D2E	3900	9	0.10	55.0	2R5TPE220M9	3000	3	
	105	2.5	105		7.3	4.3	1.8		3100	15	0.10	55.0	2R5TPE220MF	3000		
	105	2.5	105		7.3	4.3	1.8		2800	18	0.10	55.0	2R5TPE220MI	3000		
	105	2.5	105	330	7.3	4.3	1.8		4400	7	0.10	82.5	2R5TPE330M7	3000		
	105	2.5	105		7.3	4.3	1.8		3900	9	0.10	82.5	2R5TPE330M9	3000		
	105	2.5	105		7.3	4.3	1.8		3500	12	0.10	82.5	2R5TPE330MC	3000		
	105	2.5	105		7.3	4.3	1.8		3100	15	0.10	82.5	2R5TPE330MF	3000		
	105	2.5	105		7.3	4.3	1.8		2800	18	0.10	82.5	2R5TPE330MI	3000		
	105	2.5	105	470	7.3	4.3	1.8		4400	7	0.10	117.5	2R5TPE470M7	3000		
	105	2.5	105		7.3	4.3	1.8		3900	9	0.10	117.5	2R5TPE470M9	3000		
	105	2.5	105		7.3	4.3	1.8		3500	12	0.10	117.5	2R5TPE470MC	3000		
	105	2.5	105		7.3	4.3	1.8		3100	15	0.10	117.5	2R5TPE470MF	3000		
	105	2.5	105		7.3	4.3	1.8		2800	18	0.10	117.5	2R5TPE470MI	3000		
	105	2.5	105	680	7.3	4.3	2.8		3500	12	0.10	170.0	2R5TPE680MCL	2500		
	105	2.5	105		7.3	4.3	2.8		3100	15	0.10	170.0	2R5TPE680MFL	2500		
	105	2.5	105		7.3	4.3	2.8		1850	40	0.10	170.0	2R5TPE680MWL	2500		
	105	2.5	105	1000	7.3	4.3	3.8		3900	15	0.15	250.0	2R5TPE1000MF	2000		
	105	2.5	105		7.3	4.3	3.8	D4	4400	12	0.15	375.0	2R5TPE1500MC	2000	-	
	105	2.5	105		7.3	4.3	3.8		3900	15	0.15	375.0	2R5TPE1500MF	2000		
4.0	105	4.0	105	150	7.3	4.3	1.8	D2E	2800	18	0.10	60.0	4TPE150MI	3000	3	2a
	105	4.0	105		7.3	4.3	1.8		3100	15	0.10	88.0	4TPE220MF	3000		
	105	4.0	105		7.3	4.3	1.8		2800	18	0.10	88.0	4TPE220MI	3000		
	105	4.0	105	220	7.3	4.3	1.8		2400	25	0.10	88.0	4TPE220M	3000		
	105	4.0	105		7.3	4.3	1.8		1750	45	0.10	88.0	4TPE220MV	3000		
	105	4.0	105		7.3	4.3	1.8		2800	18	0.10	132.0	4TPE330MI	3000		
	105	4.0	105		7.3	4.3	1.8		2400	25	0.10	132.0	4TPE330M	3000		
	105	4.0	105		7.3	4.3	1.8		1850	40	0.10	132.0	4TPE330MW	3000		
	105	4.0	105	330	7.3	4.3	2.8		3500	12	0.10	188.0	4TPE470MCL	2500		
	105	4.0	105		7.3	4.3	2.8		3100	15	0.10	188.0	4TPE470MFL	2500		
	105	4.0	105		7.3	4.3	2.8		2800	18	0.10	188.0	4TPE470MIL	2500		
	105	4.0	105		7.3	4.3	2.8		2400	25	0.10	188.0	4TPE470ML	2500		
	105	4.0	105		7.3	4.3	2.8		1850	40	0.10	188.0	4TPE470MWL	2500		
	105	4.0	105		7.3	4.3	2.8	D3L	2800	18	0.10	188.0	4TPE470MWL	2500		
	105	4.0	105		7.3	4.3	2.8		2400	25	0.10	188.0	4TPE470MWL	2500		
6.3	105	6.3	105	100	7.3	4.3	1.8	D2E	2800	18	0.10	63.0	6TPE100MI	3000	3	2a
	105	6.3	105		7.3	4.3	1.8		2400	25	0.10	63.0	6TPE100M	3000		
	105	6.3	105		7.3	4.3	1.8		1850	40	0.10	63.0	6TPE100MW	3000		
	105	6.3	105	150	7.3	4.3	1.8		3100	15	0.10	94.5	6TPE150MF	3000		
	105	6.3	105		7.3	4.3	1.8		2800	18	0.10	94.5	6TPE150MI	3000		
	105	6.3	105		7.3	4.3	1.8		2400	25	0.10	94.5	6TPE150M	3000		
	105	6.3	105		7.3	4.3	1.8		2800	18	0.10	138.6	6TPE220MI	3000		
	105	6.3	105		7.3	4.3	1.8		2400	25	0.10	138.6	6TPE220M	3000		
	85	5.0	105	220	7.3	4.3	1.8		2400	25	0.10	138.6	6TPE220MAP	3000		
	105	6.3	105		7.3	4.3	1.8		1850	40	0.10	138.6	6TPE220MW	3000		
	105	6.3	105		7.3	4.3	2.8	D3L	2800	18	0.10	138.6	6TPE220MIL	2500	-	
	85	5.0	105		7.3	4.3	1.8		2400	25	0.10	207.9	6TPE330MAP	3000		
	105	6.3	105	330	7.3	4.3	2.8	D2E	3100	15	0.10	207.9	6TPE330MFL	2500	3	
	105	6.3	105		7.3	4.3	2.8		2800	18	0.10	207.9	6TPE330MIL	2500		
	105	6.3	105		7.3	4.3	2.8	D4	2400	25	0.10	207.9	6TPE330ML	2500		
	NRFND	85	5.0		7.3	4.3	3.8		4400	10	0.10	207.9	6TPE330MAA	2000		
	105	6.3	105		7.3	4.3	3.8	D4	3500	18	0.15	296.1	6TPE470MI	2000		
	105	6.3	105		7.3	4.3	3.8		3000	25	0.15	296.1	6TPE470M	2000		
10	105	6.3	105	470	7.3	4.3	3.8		2500	35	0.15	296.1	6TPE470MZ	2000	3	
	105	6.3	105		7.3	4.3	3.8		2350	40	0.15	296.1	6TPE470MW	2000		
	105	6.3	105		7.3	4.3	3.8		3500	18	0.15	428.4	6TPE680MI	2000		
	105	6.3	105		7.3	4.3	3.8		3000	25	0.15	428.4	6TPE680M	2000		
	NRFND	105	10		7.3	4.3	1.8	D2E	2400	25	0.10	68.0	10TPE68M	3000	-	
	NRFND	105	10		7.3	4.3	1.8		1850	40	0.10	68.0	10TPE68MW	3000		
	105	10	105	150	7.3	4.3	2.8	D3L	2400	25	0.10	150.0	10TPE150MPL	2500	3	
	105	10	105		7.3	4.3	2.8		2800	18	0.10	220.0	10TPE220MIL	2500		
	105	10	105	220	7.3	4.3	2.8		2400	25	0.10	220.0	10TPE220ML	2500	-	2a
	105	10	105		7.3	4.3	2.8		3000	25	0.10	330.0	10TPE330M	2000		

*1: Ripple current (100 kHz / +45 °C)

*2: ESR (100 kHz / +20 °C)

*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

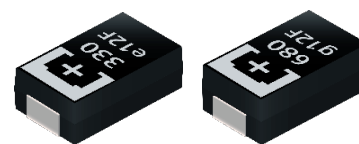
◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".

◆ Small order quantity (500 pcs/reel) is available with TPE series. Please contact our sales representative if you prefer it.

NRFND Not recommended for new design



Conductive Polymer Tantalum Solid Capacitors Surface Mount Type TPF series



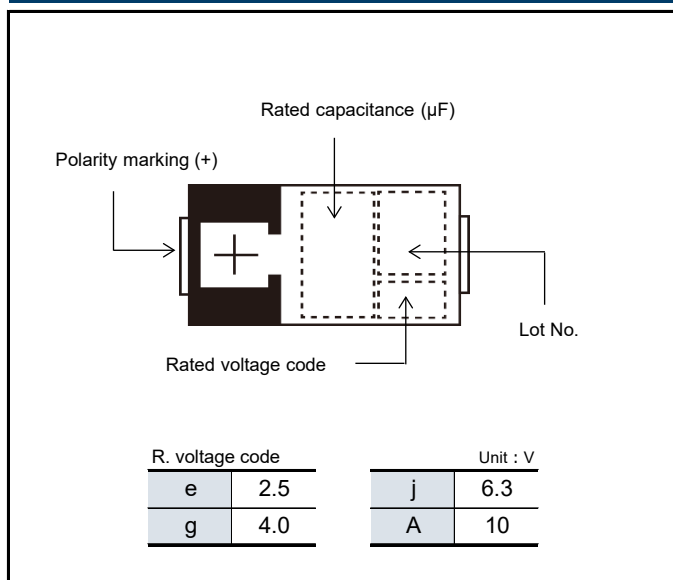
Features

- Super low ESR (5 mΩ max.)
- Large capacitance (1000 μF max.)
- RoHS compliance, Halogen free

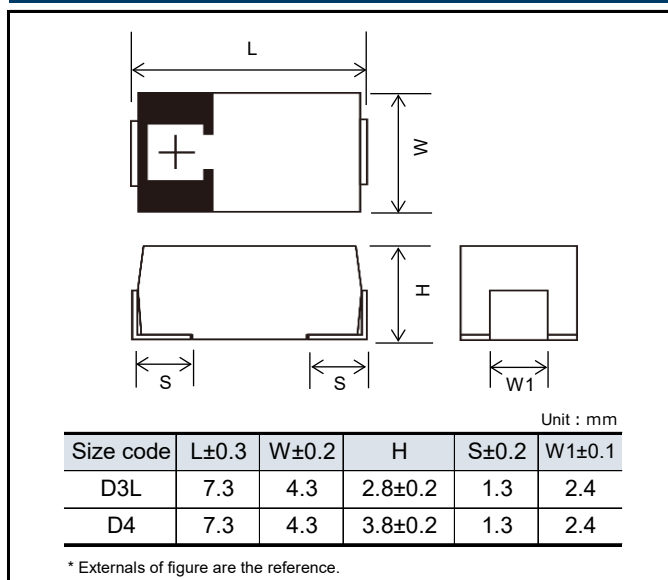
Specifications

Size code	D3L	D4
Category temp. range	-55 °C to +105 °C	
Rated voltage range	2.5 V to 10 V	2.5 V to 6.3 V
Category voltage range	2.5 V to 10 V	2.5 V to 6.3 V
Rated capacitance range	150 μF to 680 μF	470 μF to 1000 μF
Capacitance tolerance	±20 % (120 Hz / +20 °C)	
Leakage current	Please see the attached characteristics list	
Dissipation factor(tan δ)	Please see the attached characteristics list	
Surge voltage (V)	Rated voltage × 1.15	
Endurance	+105 °C 2000 h rated voltage applied	
	Capacitance change	Within ±20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	Within the initial limit
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage	
	Capacitance change	Within +50 %, -20 % of the initial value (2TPF220M6, 2TPF330M6, ETPF1000M6H (5H))
	Dissipation factor(tan δ)	Within +40 %, -20 % of the initial value (Except for above model)
	Leakage current	≤ 3 times of the initial limit

Marking



Dimensions (not to scale)



Characteristics list

Rated voltage (V)	Rated temper- ature (°C)	Cate- gory voltage (V)	Cate- gory temper- ature (°C)	Rated capaci- tance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level		
					L	W	H		Ripple ^{*1} current (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (μA)	Part number	Min. packaging q'ty (pcs)	Reflow temp ≤260°C	Reflow temp ≤250°C	
2.5	NRFND	105	2.5	105	470	7.3	4.3	2.8	D3L	4400	6	0.10	117.5	2R5TPF470M6L	2500	3	2a
		105	2.5	105		7.3	4.3	2.8		4400	7	0.10	117.5	2R5TPF470M7L	2500		
		105	2.5	105		7.3	4.3	2.8		4400	10	0.10	117.5	2R5TPF470ML	2500		
		105	2.5	105		7.3	4.3	3.8	D4	6100	5	0.10	117.5	ETPF470M5H	2000		
		105	2.5	105	680	7.3	4.3	2.8	D3L	3850	9	0.10	117.5	2R5TPF470M9L	2500		
		105	2.5	105		7.3	4.3	2.8		4400	6	0.10	170.0	2R5TPF680M6L	2500		
		105	2.5	105		7.3	4.3	2.8		4400	7	0.10	170.0	2R5TPF680M7L	2500		
		105	2.5	105		7.3	4.3	2.8		4400	10	0.10	170.0	2R5TPF680ML	2500		
		105	2.5	105	1000	7.3	4.3	3.8	D4	2700	25	0.10	170.0	ETPF680MPH	2000		
		105	2.5	105		7.3	4.3	3.8		6100	5	0.10	250.0	ETPF1000M5H	2000		
		105	2.5	105		7.3	4.3	3.8		5600	6	0.10	250.0	ETPF1000M6H	2000		
		105	2.5	105		7.3	4.3	3.8									
4.0	105	4.0	105	330	7.3	4.3	2.8	D3L	3900	9	0.10	132.0	4TPF330M9L	2500			
	105	4.0	105		7.3	4.3	2.8		4000	12	0.10	132.0	4TPF330ML	2500			
	105	4.0	105		7.3	4.3	2.8		3550	15	0.10	132.0	4TPF330MFL	2500			
	105	4.0	105	470	7.3	4.3	2.8	D4	4400	10	0.10	188.0	4TPF470ML	2500			
	105	4.0	105	680	7.3	4.3	3.8		4400	10	0.10	272.0	4TPF680MAH	2000			
	105	4.0	105		7.3	4.3	3.8		3550	15	0.10	272.0	4TPF680MFH	2000			
	105	4.0	105		7.3	4.3	3.8		2350	35	0.10	272.0	4TPF680MZH	2000			
6.3	105	6.3	105	150	7.3	4.3	2.8	D3L	2750	25	0.10	94.5	6TPF150MPL	2500	—	3	
	NRFND	105	6.3	105	220	7.3	4.3		2.8	5550	6	0.10	138.6	6TPF220M6L	2500		
	NRFND	105	6.3	105		7.3	4.3		2.8	4600	9	0.10	138.6	6TPF220M9L	2500		
	105	6.3	105	7.3		4.3	2.8		4000	12	0.10	138.6	6TPF220ML	2500			
	105	6.3	105	7.3		4.3	2.8		3550	15	0.10	138.6	6TPF220MFL	2500			
	105	6.3	105	330	7.3	4.3	2.8		2750	25	0.10	138.6	6TPF220MPL	2500			
	105	6.3	105		7.3	4.3	2.8		3900	9	0.10	207.9	6TPF330M9L	2500			
	105	6.3	105		7.3	4.3	2.8		3650	10	0.10	207.9	6TPF330MAL	2500			
	105	6.3	105		7.3	4.3	2.8		4400	10	0.10	296.1	6TPF470MAH	2000			
	105	6.3	105	470	7.3	4.3	3.8	D4	3550	15	0.10	296.1	6TPF470MFH	2000			
10	105	10	105	150	7.3	4.3	2.8	D3L	3600	15	0.10	150.0	10TPF150ML	2500	—		

*1: Ripple current (100 kHz / +45 °C)

*2: ESR (100 kHz / +20 °C)

*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

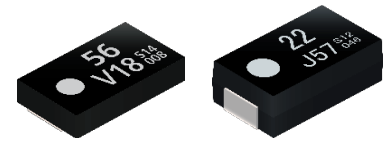
NRFND

Not recommended for new design

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".

◆ Small order quantity (500 pcs/reel) is available with TPF series. Please contact our sales representative if you prefer it.

Conductive Polymer Tantalum Solid Capacitors Surface Mount Type TQT series



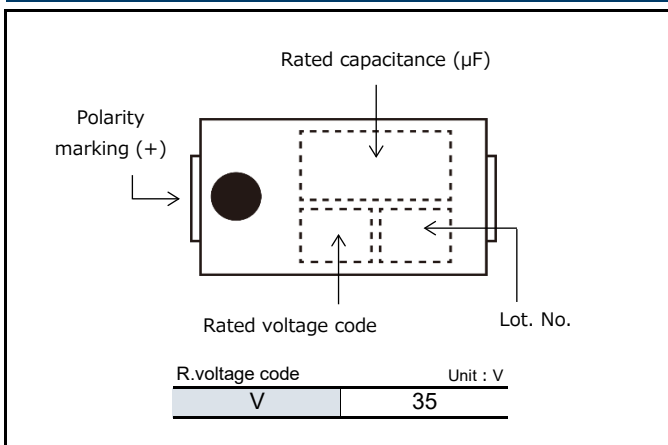
Features

- High voltage (35 V max.)
- Large capacitance
- RoHS compliance, Halogen free

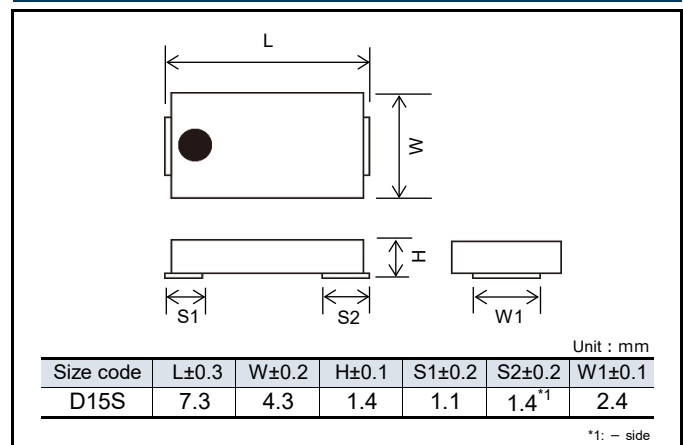
Specifications

Size code	D15S	
Category temp. range	-55 °C ~ +105 °C	
Rated volt. range	35 V	
Category volt. range	35 V	
Rated cap. range	56 µF	
Capacitance tolerance	±20 % (120 Hz / +20 °C)	
Leakage current	Please see the attached characteristics list	
Dissipation factor(tan δ)	Please see the attached characteristics list	
Surge voltage (V)	Rated voltage × 1.15	
Endurance	+105 °C 2000 h rated voltage applied	
	Capacitance change	Within ±20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	Within the initial limit
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage	
	Capacitance change	Within +40 %, -20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	≤ 3 times of the initial limit

Marking



Dimensions (not to scale)



Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (µF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	Re-flow soldering
					L	W	H		Ripple current ^{*1} (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (µA)	Part number	Min. packaging q'ty (pcs)		
35	105	35	105	56	7.3	4.3	1.4	D15S	1200	100	0.10	196	35TQT56M	4000	3	A

*1: Ripple current (100 kHz / +105 °C)

*2: ESR (100 kHz / +20 °C)

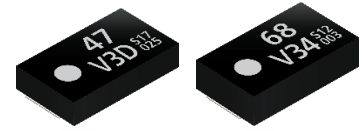
*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".



Conductive Polymer Tantalum Solid Capacitors Surface Mount Type TQS series



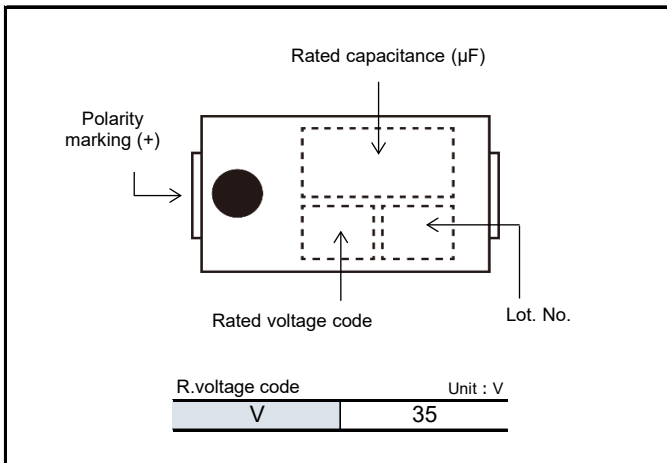
Features

- High voltage (35 V max.)
- RoHS compliance, Halogen free

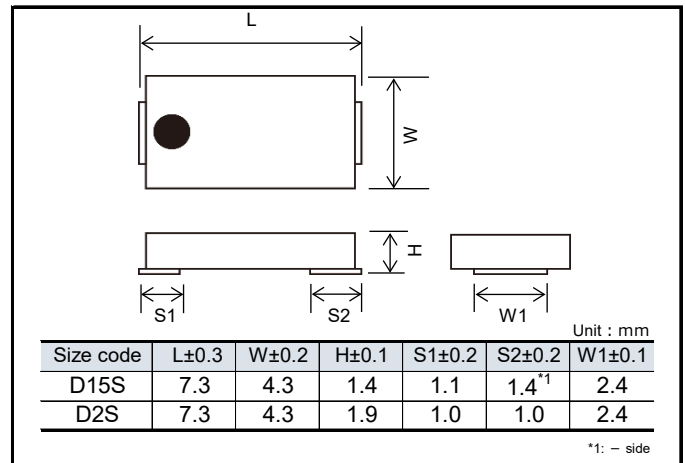
Specifications

Size code	D15S	D2S
Category temp. range	-55 °C to +105 °C	
Rated voltage range	35 V	
Category voltage range	35 V	
Rated capacitance range	47 µF	68 µF
Capacitance tolerance	±20 % (120 Hz / +20 °C)	
Leakage current	Please see the attached characteristics list	
Dissipation factor(tan δ)	Please see the attached characteristics list	
Surge voltage (V)	Rated voltage × 1.15	
Endurance	+105 °C 2000 h rated voltage applied	
	Capacitance change	Within ±20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	Within the initial limit
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage	
	Capacitance change	Within +40 %, -20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	≤ 3 times of the initial limit

Marking



Dimensions (not to scale)



Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (µF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	Re-flow soldering
					L	W	H		Ripple ^{*1} current (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (µA)	Part number	Min. packaging q'ty (pcs)		
35	105	35	105	47	7.3	4.3	1.4	D15S	1200	100	0.10	164.5	35TQS47MEU	4000	3	A
	105	35	105	47	7.3	4.3	1.4	D15S	1200	100	0.10	164.5	35TQS47MEX	4000		
	105	35	105	68	7.3	4.3	1.9	D2S	1200	100	0.10	238	35TQS68ME2	3000		

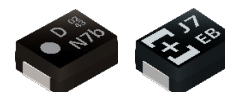
*1: Ripple current (100 kHz / +105 °C)

*2: ESR (100 kHz / +20 °C)

*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level". NRFND Not recommended for new design



Conductive Polymer Tantalum

Solid Capacitors

Surface Mount Type

TQC series

B size

Features

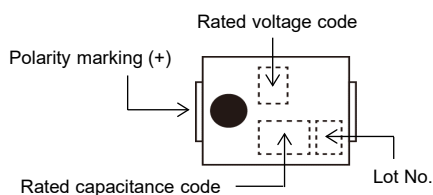
- High voltage (35 V max.)
- RoHS compliance, Halogen free

Specifications

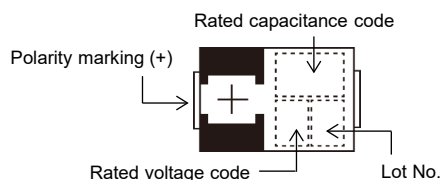
Size code	B2	
Category temp. range	-55 °C to +105 °C	
Rated voltage range	16 V to 35 V	
Category voltage range	16 V to 35 V	
Rated capacitance range	10 µF to 47 µF	
Capacitance tolerance	±20 % (120 Hz / +20 °C)	
Leakage current	Please see the attached characteristics list	
Dissipation factor(tan δ)	Please see the attached characteristics list	
Surge voltage (V)	Rated voltage × 1.15	
Endurance	+105 °C 2000 h (16TQC33MYFB : 1000 h), rated voltage applied	
	Capacitance change	Within ±20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	Within the initial limit
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage	
	Capacitance change	Within +40 %, -20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	≤ 3 times of the initial limit

Marking

〈 Laser marking 〉



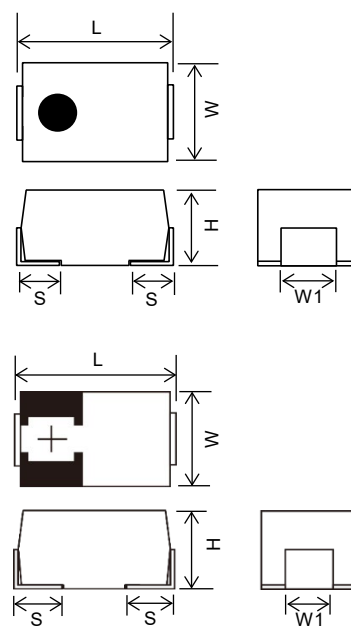
〈 Other than the above 〉



R. voltage code		Unit : V	
C	16	E	25
D	20	V	35

R. capacitance code		Unit : µF	
A7	10	J7	22
E7	15	N7	33
		S7	47

Dimensions (not to scale)



Unit : mm					
Size code	L±0.2	W±0.2	H±0.1	S±0.2	W1±0.1
B2	3.5	2.8	1.9	0.8	2.2

* Externals of figure are the reference.

Characteristics list

Rated voltage (V)	Rated temper- ature (°C)	Cate- gory voltage (V)	Cate- gory temper- ature (°C)	Rated capaci- tance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	Re- flow solder- ing	
					L	W	H		Ripple ^{*1} current (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (μA)	Part number	Min. packaging q'ty (pcs)			
16	NRFND	105	16	105	15	3.5	2.8	1.9	B2	1000	90	0.10	72.0	16TQC15M	2000	3	D
	NRFND	105	16	105	22	3.5	2.8	1.9		1000	90	0.10	35.2	16TQC22MYFB	2000		
	NRFND	105	16	105	33	3.5	2.8	1.9		1000	90	0.10	158.4	16TQC33MYFB	2000		
		105	16	105		3.5	2.8	1.9		1000	90	0.10	158.4	16TQC33MB2	2000		
		105	16	105	47	3.5	2.8	1.9		1000	90	0.15	225.6	16TQC47MYFB	2000		
20	NRFND	105	20	105	22	3.5	2.8	1.9		1000	90	0.10	132.0	20TQC22MYFB	2000	3	D
		105	20	105		3.5	2.8	1.9		1000	90	0.10	132.0	20TQC22MB2	2000		
		105	20	105	33	3.5	2.8	1.9		900	100	0.10	198.0	20TQC33MYFB	2000		
25	NRFND	105	25	105	15	3.5	2.8	1.9		900	100	0.10	25.0	25TQC10MEB	2000		D
	NRFND	105	25	105		3.5	2.8	1.9		900	100	0.10	112.5	25TQC15MYFB	2000		
		105	25	105		3.5	2.8	1.9		900	100	0.10	112.5	25TQC15MB	2000		
		105	25	105	22	3.5	2.8	1.9		1100	100	0.10	165.0	25TQC22MYFB	2000	A	
35		105	35	105	10	3.5	2.8	1.9		900	150	0.15	105.0	35TQC10MYFB	2000		
		105	35	105		3.5	2.8	1.9		750	200	0.15	105.5	35TQC10MXB	2000		

*1: Ripple current (100 kHz / +105 °C)

*2: ESR (100 kHz / +20 °C)

*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

NRFND Not recommended for new design

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".

◆ Small order quantity (500 pcs/reel) is available with TQC series. Please contact our sales representative if you prefer it.



Conductive Polymer Tantalum Solid Capacitors

Surface Mount Type

TQC series D size



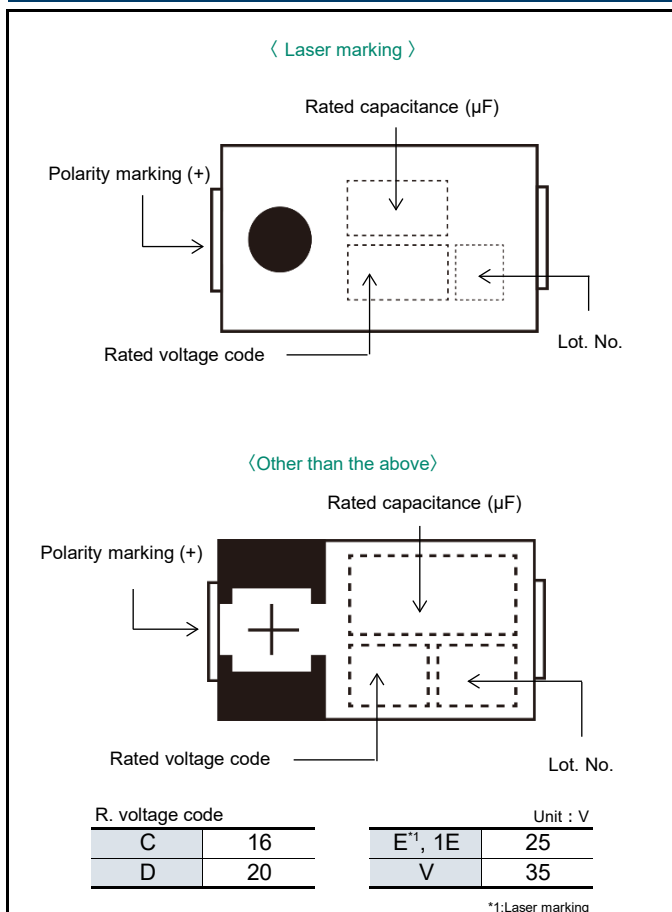
Features

- High voltage (35 V max.)
- RoHS compliance, Halogen free

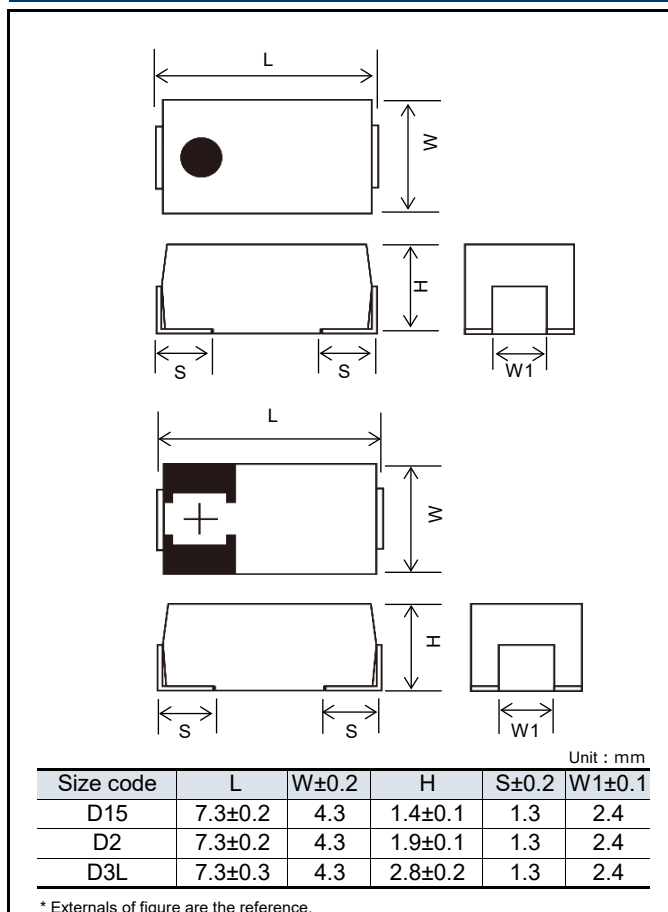
Specifications

Size code	D15	D2	D3L
Category temp. range	-55 °C to +105 °C		
Rated voltage range	16 V to 25 V	16 V to 35 V	16 V to 25 V
Category voltage range	16 V to 25 V	16 V to 35 V	16 V to 25 V
Rated capacitance range	22 μF to 47 μF	10 μF to 150 μF	68 μF to 220 μF
Capacitance tolerance	±20 % (120 Hz / +20 °C)		
Leakage current	Please see the attached characteristics list		
Dissipation factor(tan δ)	Please see the attached characteristics list		
Surge voltage (V)	Rated voltage × 1.15		
Endurance	+105 °C 2000 h rated voltage applied		
	Capacitance change	Within ±20 % of the initial value	
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit	
	Leakage current	Within the initial limit	
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage		
	Capacitance change	Within +40 %, -20 % of the initial value	
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit	
	Leakage current	≤ 3 times of the initial limit	

Marking



Dimensions (not to scale)



Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	Re-flow soldering
					L	W	H		Ripple ^{*1} current (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (μA)	Part number	Min. packaging q'ty (pcs)		
16	105	16	105	47	7.3	4.3	1.4	D15	1500	55	0.10	75.2	16TQC47MYFT	3000	3	D
	105	16	105		7.3	4.3	1.9	D2	1450	55	0.10	75.2	16TQC47MD2	3000		A
	105	16	105		7.3	4.3	1.9		1250	70	0.10	75.2	16TQC47MBD2	3000		
	105	16	105	68	7.3	4.3	1.9	D2	1500	50	0.10	108.8	16TQC68MD2	3000		A
	105	16	105	100	7.3	4.3	1.9		1800	50	0.10	160.0	16TQC100MD2	3000		
	105	16	105	150	7.3	4.3	1.9		1500	70	0.15	240.0	1CTQC15173F1	3000		D
	105	16	105		7.3	4.3	2.8	D3L	1800	50	0.10	240.0	16TQC150MD3	2500		B
	105	16	105		7.3	4.3	2.8		1750	50	0.10	352.0	16TQC220MD3	2500		C
20	105	20	105	22	7.3	4.3	1.9	D2	1200	90	0.10	44.0	20TQC22MQD	3000	3	D
	105	20	105	33	7.3	4.3	1.9		1400	60	0.10	66.0	20TQC33MD2	3000		A
	105	20	105	47	7.3	4.3	1.4	D15	1500	55	0.10	94.0	20TQC47MYFT	3000		D
	105	20	105		7.3	4.3	1.9	D2	1450	55	0.10	94.0	20TQC47MD2	3000		A
	105	20	105		7.3	4.3	1.9		1250	100	0.15	200.0	20TQC100MD2	3000		D
	105	20	105	100	7.3	4.3	2.8	D3L	1700	55	0.10	200.0	20TQC100MD3	2500		B
25	105	25	105	15	7.3	4.3	1.9	D2	1000	90	0.10	37.5	25TQC15MD2	3000	3	A
	105	25	105	22	7.3	4.3	1.9		1400	60	0.10	55.0	25TQC22MD2	3000		
	105	25	105	33	7.3	4.3	1.9		1400	60	0.10	82.5	25TQC33MD2	3000		
	105	25	105	68	7.3	4.3	2.8	D3L	1400	70	0.10	170.0	25TQC68MD3	2500		B
	105	25	105	100	7.3	4.3	2.8		1600	60	0.10	250.0	25TQC100MD3	2500		C
35	105	35	105	10	7.3	4.3	1.9	D2	820	180	0.10	35.0	35TQC10MD2	3000	3	A
	105	35	105	15	7.3	4.3	1.9		900	150	0.10	52.5	35TQC15MD2	3000		
	105	35	105	22	7.3	4.3	1.9		900	150	0.10	77.0	35TQC22MD2	3000		

NRFND [Not recommended for new design]

16	105	16	105	33	7.3	4.3	1.9	D2	1400	70	0.10	52.8	16TQC33MYFD	3000	3	D
	105	16	105	47	7.3	4.3	1.9		1450	55	0.10	75.2	16TQC47MYFD	3000		
	105	16	105		7.3	4.3	1.9		1250	70	0.10	75.2	16TQC47MBD	3000		
	105	16	105	68	7.3	4.3	1.9		1500	50	0.10	108.8	16TQC68MYF	3000		
	105	16	105	100	7.3	4.3	1.9		1800	50	0.10	160.0	16TQC100MYF	3000		
	105	16	105	150	7.3	4.3	2.8	D3L	1800	50	0.10	240.0	16TQC150MYF	2500		
20	105	20	105	33	7.3	4.3	1.9	D2	1400	60	0.10	66.0	20TQC33MYFD	3000		
	105	20	105	47	7.3	4.3	1.9		1450	55	0.10	94.0	20TQC47MYF	3000		
	105	20	105	100	7.3	4.3	2.8	D3L	1700	55	0.10	200.0	20TQC100MYF	2500		
25	105	25	105	15	7.3	4.3	1.9	D2	1000	90	0.10	38.0	25TQC15MYFD	3000		
	105	25	105	22	7.3	4.3	1.9		1400	60	0.10	55.0	25TQC22MYFD	3000		
	105	25	105		7.3	4.3	1.4	D15	1400	70	0.10	55.0	25TQC22MYFT	3000		
	105	25	105		7.3	4.3	1.9	D2	1400	60	0.10	82.5	25TQC33MYF	3000		
	105	25	105	68	7.3	4.3	2.8	D3L	1400	70	0.10	170.0	25TQC68MYF	2500		
35	105	35	105	10	7.3	4.3	1.9	D2	1000	120	0.10	35.0	35TQC10MYF	3000		
	105	35	105	15	7.3	4.3	1.9		900	150	0.10	52.5	35TQC15MYF	3000		

*1: Ripple current (100 kHz / +105 °C)

*2: ESR (100 kHz / +20 °C)

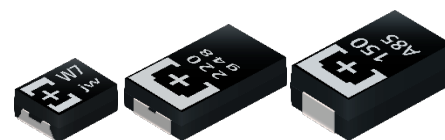
*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".

◆ Small order quantity (500 pcs/reel) is available with TQC series. Please contact our sales representative if you prefer it.

Conductive Polymer Tantalum Solid Capacitors Surface Mount Type TA series



■ This product is not intended for use in any driving application or any other critical functions that affect passenger safety.(e.g. Powertrain, ABS, Engine ECU, Airbag, etc.)
If the intended use of TA/TV series products is for use in other automotive related applications, please contact our sales team. All requests are subject to approval.

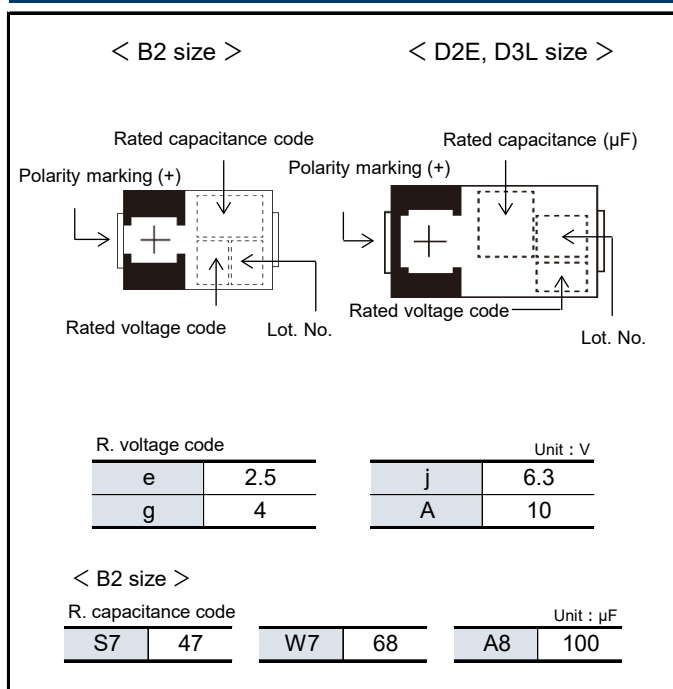
Features

- Guaranteed at 85 °C 85 %RH
- RoHS compliance, Halogen free

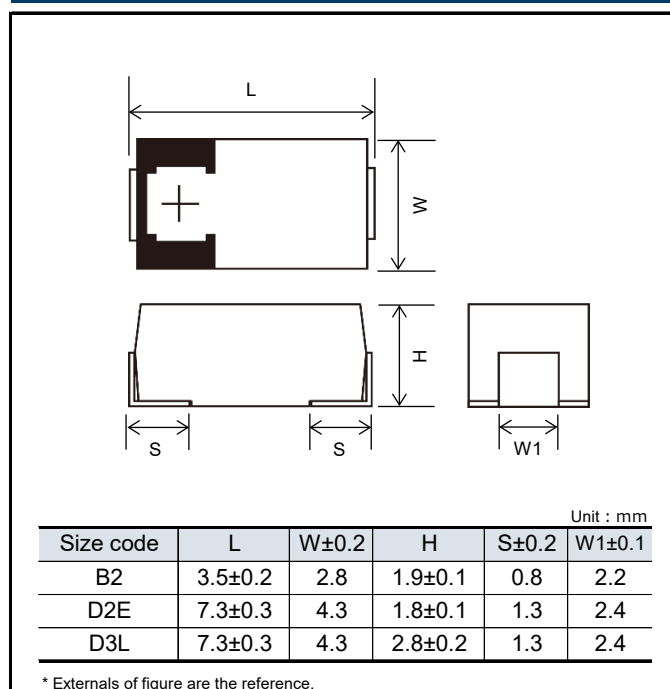
Specifications

Size code	B2	D2E	D3L
Category temp. range	-55 °C to +105 °C		
Rated voltage range	4.0 V to 10 V	2.5 V to 6.3 V	2.5 V to 10 V
Category voltage range	4.0 V to 10 V	2.5 V to 6.3 V	2.5 V to 10 V
Rated capacitance range	47 μF to 100 μF	150 μF to 470 μF	150 μF to 680 μF
Capacitance tolerance	±20 % (120 Hz / +20 °C)		
Leakage current	Please see the attached characteristics list		
Dissipation factor(tan δ)	Please see the attached characteristics list		
Surge voltage (V)	Rated voltage × 1.15		
Endurance	+105 °C 2000 h (B2 size : 1000 h) , rated voltage applied		
	Capacitance change	Within ±20 % of the initial value	
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit	
	Leakage current	Within the initial limit	
Damp heat (Steady State)	+85 °C, 85 % to 90 % RH, 500 h, rated voltage applied		
	Capacitance change	Within +50 %, -20 % of the initial value (2R5TAE470M (F), 2R5TAE330M (I, F), 2R5TAE220M (F, 9))	
		Within +40 %, -20 % of the initial value (Except for above model)	
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit	
Leakage current	Within the initial limit		

Marking



Dimensions (not to scale)



Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	
					L	W	H		Ripple current ^{*1} (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (μA)	Part number	Min. packaging q'ty (pcs)	Reflow temp ≤260°C	Reflow temp ≤250°C
2.5	105	2.5	105	220	7.3	4.3	1.8	D2E	3900	9	0.10	110.0	2R5TAE220M9	3000	3	3
	105	2.5	105		7.3	4.3	1.8		3100	15	0.10	55.0	2R5TAE220MF	3000		
	105	2.5	105		7.3	4.3	1.8		2400	25	0.10	55.0	2R5TAE220M	3000		
	105	2.5	105	330	7.3	4.3	1.8		3100	15	0.10	82.5	2R5TAE330MF	3000		
	105	2.5	105		7.3	4.3	1.8		2800	18	0.10	82.5	2R5TAE330MI	3000		
	105	2.5	105		7.3	4.3	1.8		2400	25	0.10	82.5	2R5TAE330M	3000		
	105	2.5	105	470	7.3	4.3	1.8	D3L	3100	15	0.10	117.5	2R5TAE470MF	3000		
	105	2.5	105		7.3	4.3	1.8		2400	25	0.10	117.5	2R5TAE470M	3000		
	105	2.5	105		7.3	4.3	2.8		3100	15	0.10	170.0	2R5TAE680MFL	2500		
	105	2.5	105	680	7.3	4.3	2.8		2400	25	0.10	170.0	2R5TAE680ML	2500		
4.0	105	4.0	105	100	3.8	2.8	1.9	B2	1100	70	0.08	40.0	4TAB100M	2000		
	105	4.0	105	220	7.3	4.3	1.8	D2E	2800	18	0.10	88.0	4TAE220MI	3000		
	105	4.0	105		7.3	4.3	1.8		2400	25	0.10	88.0	4TAE220M	3000		
	105	4.0	105	470	7.3	4.3	2.8	D3L	2800	18	0.10	188.0	4TAE470MIL	2500		
	105	4.0	105		7.3	4.3	2.8		2400	25	0.10	188.0	4TAE470ML	2500		
6.3	105	6.3	105	47	3.5	2.8	1.9	B2	1100	70	0.08	29.6	6TAB47M	2000		
	105	6.3	105	68	3.5	2.8	1.9		1100	70	0.08	42.8	6TAB68M	2000		
	105	6.3	105	150	7.3	4.3	1.8	D2E	2400	25	0.10	94.5	6TAE150M	3000		
	105	6.3	105	220	7.3	4.3	1.8		2800	18	0.10	138.6	6TAE220MI	3000		
	105	6.3	105		7.3	4.3	1.8	D3L	2400	25	0.10	138.6	6TAE220M	3000		
	105	6.3	105	330	7.3	4.3	2.8		2400	25	0.10	207.9	6TAE330ML	2500		
10	105	10	105	47	3.5	2.8	1.9	B2	1100	70	0.08	47.0	10TAB47M	2000		
	105	10	105	150	7.3	4.3	2.8	D3L	2400	25	0.10	150.0	10TAE150ML	2500		
	105	10	105	220	7.3	4.3	2.8		2400	25	0.10	220.0	10TAE220ML	2500		

*1: Ripple current (100 kHz / +45 °C)

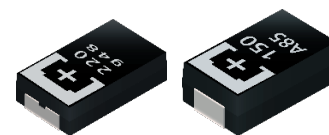
*2: ESR (100 kHz / +20 °C)

*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".

Conductive Polymer Tantalum Solid Capacitors Surface Mount Type TV series



■ This product is not intended for use in any driving application or any other critical functions that affect passenger safety (e.g. Powertrain, ABS, Engine ECU, Airbag, etc.)
If the intended use of TA/TV series products is for use in other automotive related applications, please contact our sales team. All requests are subject to approval.

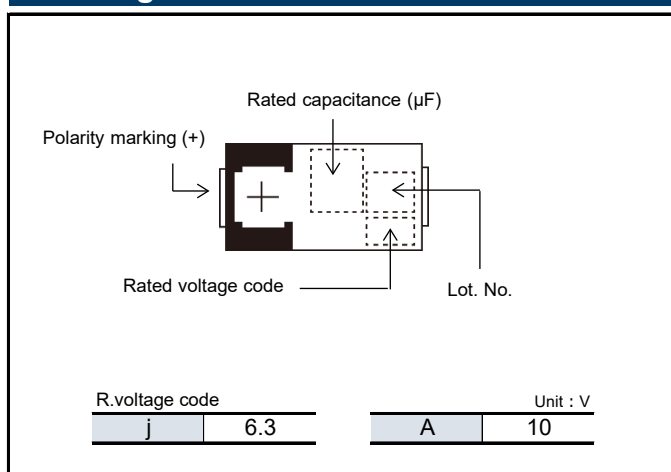
Features

- Guaranteed at 85 °C 85 %RH
- Guaranteed at 125 °C
- RoHS compliance, Halogen free

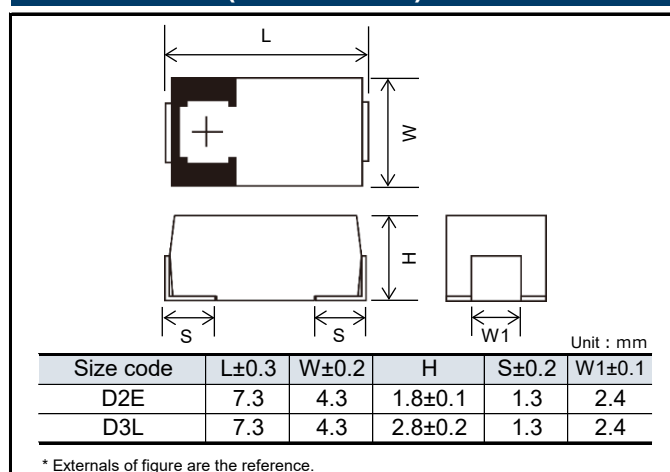
Specifications

Size code	D2E	D3L
Category temp. range	-55 °C to +125 °C	
Rated voltage range	6.3 V to 10 V	10 V
Category voltage range	4.0 V to 6.3 V	6.3 V
Rated capacitance range	68 µF to 150 µF	150 µF
Capacitance tolerance	±20 % (120 Hz / +20 °C)	
Leakage current	Please see the attached characteristics list	
Dissipation factor(tan δ)	Please see the attached characteristics list	
Surge voltage (V)	Rated voltage × 1.15	
Endurance	+125 °C 1000 h, category voltage applied (+105 °C 2000 h, rated voltage applied)	
	Temperature	125 °C
	Capacitance change	Within ±20% of the initial value
	Dissipation factor(tan δ)	≤ 2 times of the initial limit
Damp heat (Steady State)	Leakage current	≤ 2 times of the initial limit
	Capacitance change	Within ±20% of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	Within the initial limit

Marking



Dimensions (not to scale)



Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (µF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	
					L	W	H		Ripple current*1 (mA rms)	ESR*2 (mΩ max.)	tan δ*3	LC*4 (µA)	Part number	Min. packaging q'ty (pcs)	Reflow temp ≤260°C	Reflow temp ≤250°C
6.3	105	4.0	125	150	7.3	4.3	1.8	D2E	2400	25	0.10	94.5	6TVE150M	3000	5	3
10	105	6.3	125	68	7.3	4.3	1.8	D2E	2400	25	0.10	68.0	10TVE68M	3000		
	105	6.3	125	150	7.3	4.3	2.8	D3L	2400	25	0.10	150.0	10TVE150ML	2500		

*1: Ripple current (100 kHz / +45 °C)

*2: ESR (100 kHz / +20 °C)

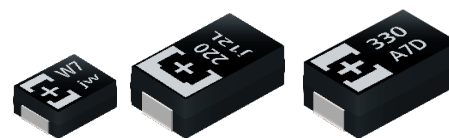
*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".



Conductive Polymer Tantalum Solid Capacitors Surface Mount Type TPB series



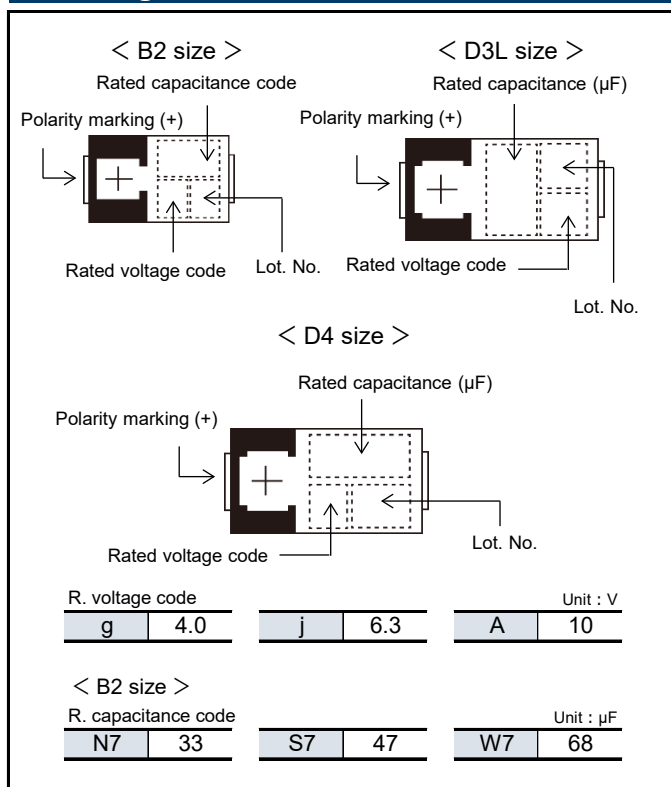
Features

- Standard
- RoHS compliance, Halogen free

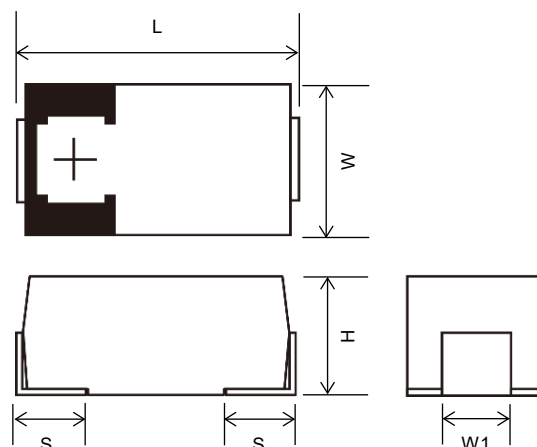
Specifications

Size code	B2	D3L	D4
Category temp. range	-55 °C to +105 °C		
Rated voltage range	4.0 V to 10 V	4.0 V to 10 V	6.3 V to 10 V
Category voltage range	4.0 V to 10 V	4.0 V to 10 V	6.3 V to 10 V
Rated capacitance range	33 μF to 68 μF	150 μF to 330 μF	220 μF to 470 μF
Capacitance tolerance	±20 % (120 Hz / +20 °C)		
Leakage current	Please see the attached characteristics list		
Dissipation factor(tan δ)	Please see the attached characteristics list		
Surge voltage (V)	Rated voltage × 1.15		
Endurance	+105 °C 2000 h, (B2 size : 1000 h) rated voltage applied		
	Rated temp. +85 °C 1000 h rated voltage applied		
	Capacitance change	Within ±20 % of the initial value	
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit	
	Leakage current	Within the initial limit	
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage		
	Capacitance change	Within +40 %, -20 % of the initial value (Except for above model)	
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit	
	Leakage current	≤ 3 times of the initial limit	

Marking



Dimensions (not to scale)



Size code	L	W±0.2	H	S±0.2	W1±0.1
B2	3.5±0.2	2.8	1.9±0.1	0.8	2.2
D3L	7.3±0.3	4.3	2.8±0.2	1.3	2.4
D4	7.3±0.3	4.3	3.8±0.2	1.3	2.4

* Externals of figure are the reference.

Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	
					L	W	H		Ripple current ^{*1} (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (μA)	Part number	Min. packaging q'ty (pcs)	Reflow temp ≤260°C	Reflow temp ≤250°C
4.0	105	4.0	105	68	3.5	2.8	1.9	B2	1100	70	0.08	27.2	4TPB68M	2000		3
	NRFND 105	4.0	105	330	7.3	4.3	2.8	D3L	2000	40	0.10	132	4TPB330ML	2500		2a
6.3	105	6.3	105	33	3.5	2.8	1.9	B2	1100	70	0.08	20.7	6TPB33M	2000	3	3
	105	6.3	105	68	3.5	2.8	1.9		1100	70	0.08	42.8	6TPB68M	2000		
	NRFND 105	6.3	105	220	7.3	4.3	2.8	D3L	2000	40	0.10	138.6	6TPB220ML	2500		2a
	NRFND 105	6.3	105		7.3	4.3	2.8		1750	50	0.10	138.6	6TPB220MTL	2500		
	85	5.0	105	330	7.3	4.3	2.8		2000	40	0.10	207.9	6TPB330MAL	2500		
	105	6.3	105		7.3	4.3	2.8		2000	40	0.10	207.9	6TPB330ML	2500		
	105	6.3	105		7.3	4.3	3.8	D4	3000	40	0.10	207.9	6TPB330M	2000		
	105	6.3	105		7.3	4.3	2.8	D3L	1850	45	0.10	207.9	6TPB330MVL	2500		
	105	6.3	105	470	7.3	4.3	3.8	D4	3000	35	0.15	296.1	6TPB470M	2000		
10	105	10	105	33	3.5	2.8	1.9	B2	1100	70	0.08	33.0	10TPB33M	2000		3
	105	10	105	47	3.5	2.8	1.9		1100	70	0.08	47.0	10TPB47M	2000		
	105	10	105	150	7.3	4.3	2.8	D3L	2000	40	0.10	150.0	10TPB150ML	2500		2a
	105	10	105		7.3	4.3	2.8		1700	55	0.10	150.0	10TPB150MGL	2500		
	105	10	105	220	7.3	4.3	2.8		2000	40	0.10	220.0	10TPB220ML	2500		
	105	10	105		7.3	4.3	3.8		3000	40	0.10	220.0	10TPB220M	2000		
	NRFND 105	10	105	330	7.3	4.3	3.8	D4	3000	35	0.10	330.0	10TPB330M	2000		
	105	10	105		7.3	4.3	3.8		2800	40	0.10	330.0	10TPB330MW	2000		

*1: Ripple current (100 kHz / +45 °C)

NRFND Not recommended for new design

*2: ESR (100 kHz / +20 °C)

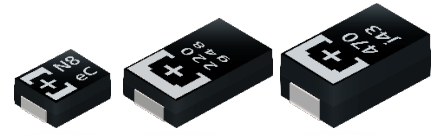
*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".



Conductive Polymer Tantalum Solid Capacitors Surface Mount Type TC series



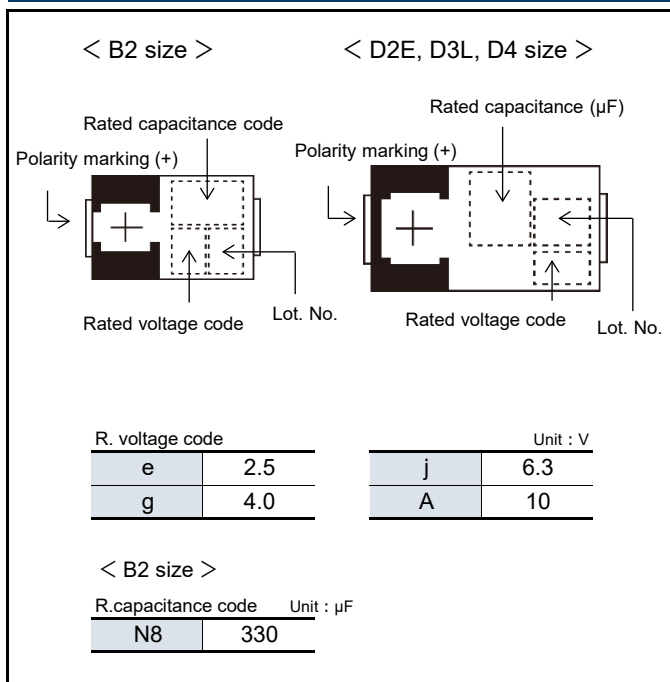
Features

- Guaranteed at 125 °C 1000 h
- RoHS compliance, Halogen free

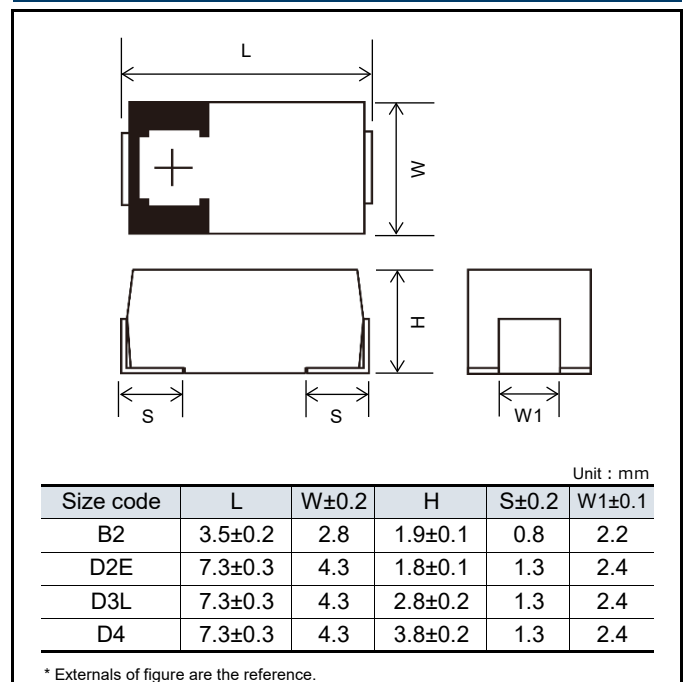
Specifications

Size code	B2	D2E	D3L	D4
Category temp. range	-55 °C to +125 °C			
Rated voltage range	2.5 V	4.0 V to 6.3 V	2.5 V to 10 V	
Category voltage range	2.0 V	3.2 V to 5.0 V	2.0 V to 8.0 V	
Rated capacitance range	330 μF	100 μF to 330 μF	150 μF to 680 μF	330 μF to 1000 μF
Capacitance tolerance	±20 % (120 Hz / +20 °C)			
Leakage current	Please see the attached characteristics list			
Dissipation factor(tan δ)	Please see the attached characteristics list			
Surge voltage (V)	Rated voltage × 1.15			
Endurance	+125 °C 1000 h, category voltage applied			
	Capacitance change	Within ±20 % of the initial value		
	Dissipation factor(tan δ)	≤ 2 times of the initial limit		
	Leakage current	≤ 2 times of the initial limit		
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage			
	Capacitance change	Within +50 %, -20 % of the initial value (ETCF1000M6H(5H))		
		Within +40 %, -20 % of the initial value		
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit		
Leakage current	≤ 3 times of the initial limit			

Marking



Dimensions (not to scale)



Characteristics list

Series	Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temp. (°C)	Rated capacitance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	
						L	W	H		Ripple ^{*1} current (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (μA)	Part number	Min. packaging q'ty (pcs)	Reflow temp ≤260°C	Reflow temp ≤250°C
TCE	2.5	105	2.0	125	330	3.5	2.8	1.9	B2	3200	9/300kHz	0.08	165.0	ETCE330M9GB	2000	3	2a
		105	2.0	125	680	7.3	4.3	2.8	D3L	3500	12	0.10	170.0	ETCE680MCL	2500		
		105	2.0	125		7.3	4.3	2.8		3100	15	0.10	170.0	ETCE680MFL	2500		
		105	2.0	125	1000	7.3	4.3	3.8	D4	3900	15	0.15	250.0	ETCE1000MF	2000		
	4.0	105	3.2	125	150	7.3	4.3	1.8	D2E	2800	18	0.10	60.0	4TCE150MI	3000		
		105	3.2	125	220	7.3	4.3	1.8		3100	15	0.10	88.0	4TCE220MF	3000		
		105	3.2	125		7.3	4.3	1.8		2800	18	0.10	88.0	4TCE220MI	3000		
		105	3.2	125	330	7.3	4.3	1.8		2400	25	0.10	88.0	4TCE220M	3000		
		105	3.2	125		7.3	4.3	1.8	D3L	2800	18	0.10	132.0	4TCE330MI	3000		
		105	3.2	125		7.3	4.3	1.8		2400	25	0.10	132.0	4TCE330M	3000		
		105	3.2	125		7.3	4.3	2.8		3500	12	0.10	188.0	4TCE470MCL	2500		
		105	3.2	125	470	7.3	4.3	2.8		3100	15	0.10	188.0	4TCE470MFL	2500		
		105	3.2	125		7.3	4.3	2.8		2800	18	0.10	188.0	4TCE470MIL	2500		
		105	3.2	125		7.3	4.3	2.8		2400	25	0.10	188.0	4TCE470ML	2500		
		105	3.2	125		7.3	4.3	2.8		2400	25	0.10	188.0	4TCE470ML	2500		
	6.3	105	5.0	125	100	7.3	4.3	1.8	D2E	2800	18	0.10	63.0	6TCE100MI	3000		
		105	5.0	125		7.3	4.3	1.8		2400	25	0.10	63.0	6TCE100M	3000		
		105	5.0	125	150	7.3	4.3	1.8		3100	15	0.10	94.5	6TCE150MF	3000		
		105	5.0	125		7.3	4.3	1.8		2800	18	0.10	94.5	6TCE150MI	3000		
		105	5.0	125	220	7.3	4.3	1.8	D2E	2400	25	0.15	94.5	6TCE150M	3000		
		105	5.0	125		7.3	4.3	1.8		2800	18	0.15	138.6	6TCE220MI	3000		
		105	5.0	125		7.3	4.3	1.8		2400	25	0.15	138.6	6TCE220M	3000		
		105	5.0	125		7.3	4.3	2.8	D3L	3100	15	0.10	207.9	6TCE330MFL	2500		
		105	5.0	125	330	7.3	4.3	2.8		2800	18	0.10	207.9	6TCE330MIL	2500		
		105	5.0	125		7.3	4.3	2.8		2400	25	0.10	207.9	6TCE330ML	2500		
		105	5.0	125		7.3	4.3	3.8	D4	3500	18	0.15	296.1	6TCE470MI	2000		
		105	5.0	125	470	7.3	4.3	3.8		3000	25	0.15	296.1	6TCE470M	2000		
		105	5.0	125		7.3	4.3	3.8		3500	18	0.15	428.4	6TCE680MI	2000		
		105	5.0	125		7.3	4.3	3.8		3000	25	0.15	428.4	6TCE680M	2000		
	10	105	8.0	125	220	7.3	4.3	2.8	D3L	2800	18	0.10	220.0	10TCE220MIL	2500	-	
		105	8.0	125		7.3	4.3	2.8		2400	25	0.10	220.0	10TCE220ML	2500		
		105	8.0	125	330	7.3	4.3	3.8	D4	3000	25	0.10	330.0	10TCE330M	2000		
TCF	2.5	105	2.0	125	680	7.3	4.3	2.8	D3L	4400	6	0.10	170.0	ETCF680M6L	2500	3	
		105	2.0	125		7.3	4.3	2.8		4400	7	0.10	170.0	ETCF680M7L	2500		
		105	2.0	125		7.3	4.3	2.8		4400	10	0.10	170.0	ETCF680ML	2500		
		105	2.0	125	1000	7.3	4.3	3.8	D4	6100	5	0.10	250.0	ETCF1000M5H	2000		
		105	2.0	125		7.3	4.3	3.8		5600	6	0.10	250.0	ETCF1000M6H	2000		
	4.0	105	3.2	125	330	7.3	4.3	2.8	D3L	4000	12	0.10	132.0	4TCF330ML	2500		
		105	3.2	125	470	7.3	4.3	2.8		4400	10	0.10	188.0	4TCF470ML	2500		
		105	3.2	125	680	7.3	4.3	3.8	D4	4400	10	0.10	272.0	4TCF680MAH	2000		
	6.3	105	5.0	125	220	7.3	4.3	2.8	D3L	4600	9	0.10	138.6	6TCF220M9L	2500		
		105	5.0	125		7.3	4.3	2.8		4000	12	0.10	138.6	6TCF220ML	2500		
		105	5.0	125	330	7.3	4.3	2.8	D4	3900	9	0.10	207.9	6TCF330M9L	2500		
		105	5.0	125	470	7.3	4.3	3.8		4400	10	0.10	296.1	6TCF470MAH	2000		
	10	105	8.0	125	150	7.3	4.3	2.8	D3L	3600	15	0.10	150.0	10TCF150ML	2500	-	

*1: Ripple current (100 kHz / +45 °C)

*2: ESR (100 kHz / +20 °C)

*3: tan δ (120 Hz / +20 °C)

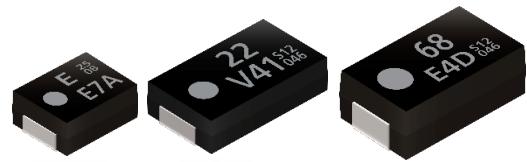
*4: After 5 minutes

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".

NRFND Not recommended for new design



Conductive Polymer Tantalum Solid Capacitors Surface Mount Type TDC series



Features

- Guaranteed at 125 °C 1000 h
- High voltage (35 V max.)
- RoHS compliance, Halogen free

Specifications

Size code	B2	D2	D3L
Category temp. range	-55 °C to +125 °C		
Rated voltage range	16 V to 25 V	16 V to 35 V	16 V to 25 V
Category voltage range	12.8 V to 20 V	12.8 V to 28 V	12.8 V to 20 V
Rated capacitance range	15 µF to 33 µF	10 µF to 100 µF	68 µF to 220 µF
Capacitance tolerance	±20 % (120 Hz / +20 °C)		
Leakage current	Please see the attached characteristics list		
Dissipation factor(tan δ)	Please see the attached characteristics list		
Surge voltage (V)	Rated voltage × 1.15		
Endurance	+125 °C 1000 h, category voltage applied		
	Capacitance change	Within ±20 % of the initial value	
	Dissipation factor(tan δ)	≤ 2 times of the initial limit	
	Leakage current	≤ 2 times of the initial limit	
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage		
	Capacitance change	Within +40 %, -20 % of the initial value	
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit	
	Leakage current	≤ 3 times of the initial limit	

Marking

〈 Laser marking (B size) 〉

〈 Laser marking (D size) 〉

〈 Other than the above 〉

Rated capacitance code (D size is capacitance (µF))

Rated voltage code

Lot No.

R. voltage code (B size)		(D size)		Unit : V
C	16	E	25	E ¹ 25 1E 35 V
D	20	D	20	

*1: Laser marking

R. capacitance code (B size)		Unit : µF	
E7	15	J7	22
N7	33		

Dimensions (not to scale)

Size code	L	W±0.2	H	S±0.2	W1±0.1	Unit : mm
B2	3.5±0.2	2.8	1.9±0.1	0.8	2.2	
D2	7.3±0.2	4.3	1.9±0.1	1.3	2.4	
D3L	7.3±0.3	4.3	2.8±0.2	1.3	2.4	

* Externals of figure are the reference.

Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	Re-flow soldering
					L	W	H		Ripple current ^{*1} (mA rms)	ESR ^{*2} (mΩ max.)	tan δ ^{*3}	LC ^{*4} (μA)	Part number	Min. packaging q'ty (pcs)		
16	105	12.8	125	33	3.5	2.8	1.9	B2	1000	90	0.10	158.4	16TDC33MB2	2000	3	A
	105	12.8	125	47	7.3	4.3	1.9	D2	1450	55	0.10	75.2	16TDC47MD2	3000		
	105	12.8	125		7.3	4.3	1.9		1250	70	0.10	75.2	16TDC47MBD2	3000		
	105	12.8	125	68	7.3	4.3	1.9		1500	50	0.10	108.8	16TDC68MD2	3000		
	105	12.8	125	100	7.3	4.3	1.9		1800	50	0.10	160.0	16TDC100MD2	3000		
	105	12.8	125	150	7.3	4.3	2.8	D3L	1800	50	0.10	240.0	16TDC150MD3	2500		B
	105	12.8	125	220	7.3	4.3	2.8		1750	50	0.10	352.0	16TDC220MD3	2500		C
20	105	16	125	22	3.5	2.8	1.9	B2	1000	90	0.10	132.0	20TDC22MB2	2000	3	A
	105	16	125	33	7.3	4.3	1.9	D2	1400	60	0.10	66.0	20TDC33MD2	3000		
	105	16	125	47	7.3	4.3	1.9		1450	55	0.10	94.0	20TDC47MD2	3000		
	105	16	125	100	7.3	4.3	2.8	D3L	1700	55	0.10	200.0	20TDC100MD3	2500		B
25	105	20	125	15	3.5	2.8	1.9	B2	900	100	0.10	112.5	25TDC15MB	2000	3	A
	105	20	125		7.3	4.3	1.9	D2	1000	90	0.10	37.5	25TDC15MD2	3000		
	105	20	125	22	7.3	4.3	1.9		1400	60	0.10	55.0	25TDC22MD2	3000		
	105	20	125	33	7.3	4.3	1.9		1400	60	0.10	82.5	25TDC33MD2	3000		
	105	20	125	68	7.3	4.3	2.8	D3L	1400	70	0.10	170.0	25TDC68MD3	2500		B
	105	20	125	100	7.3	4.3	2.8		1600	60	0.10	250.0	25TDC100MD3	2500		C
35	105	28	125	10	7.3	4.3	1.9	D2	820	180	0.10	35.0	35TDC10MD2	3000	3	A
	105	28	125	15	7.3	4.3	1.9		900	150	0.10	52.5	35TDC15MD2	3000		
	105	28	125	22	7.3	4.3	1.9		900	150	0.10	77.0	35TDC22MD2	3000		

NRFND [Not recommended for new design]

16	105	12.8	125	33	3.5	2.8	1.9	B2	1000	90	0.10	158.4	16TDC33MYFB	2000	3	D
	105	12.8	125	100	7.3	4.3	1.9	D2	1800	50	0.10	160.0	16TDC100MYF	3000		
	105	12.8	125	150	7.3	4.3	2.8	D3L	1800	50	0.10	240.0	16TDC150MYF	2500		
20	105	12.8	125	22	3.5	2.8	1.9	B2	1000	90	0.10	132.0	20TDC22MYFB	2000		
	105	12.8	125	100	7.3	4.3	2.8	D3L	1700	55	0.10	200.0	20TDC100MYF	2500		
25	105	12.8	125	15	3.5	2.8	1.9	B2	900	100	0.10	112.5	25TDC15MYFB	2000		
	105	12.8	125	68	7.3	4.3	2.8	D3L	1400	70	0.10	170.0	25TDC68MYF	2500		

*1: Ripple current (100 kHz / +105 °C)

*2: ESR (100 kHz / +20 °C)

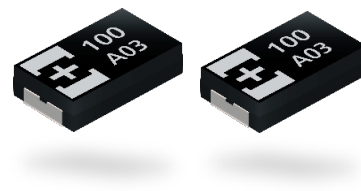
*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".



Conductive Polymer Tantalum Solid Capacitors Surface Mount Type TPC series



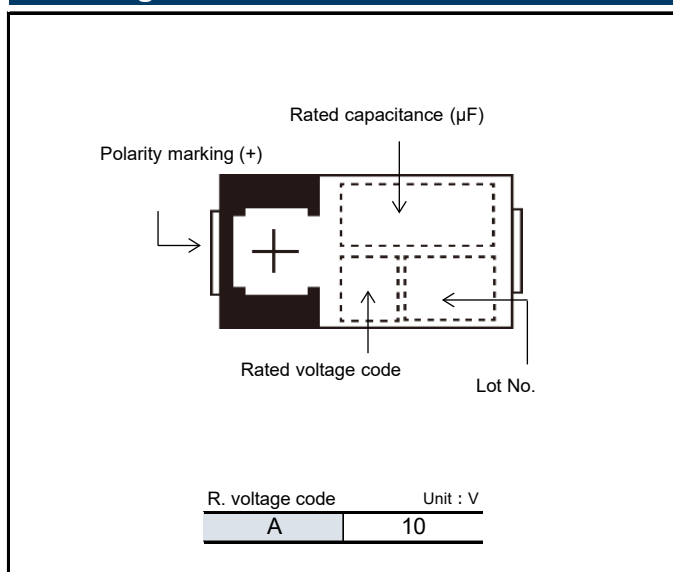
Features

- RoHS compliance, Halogen free

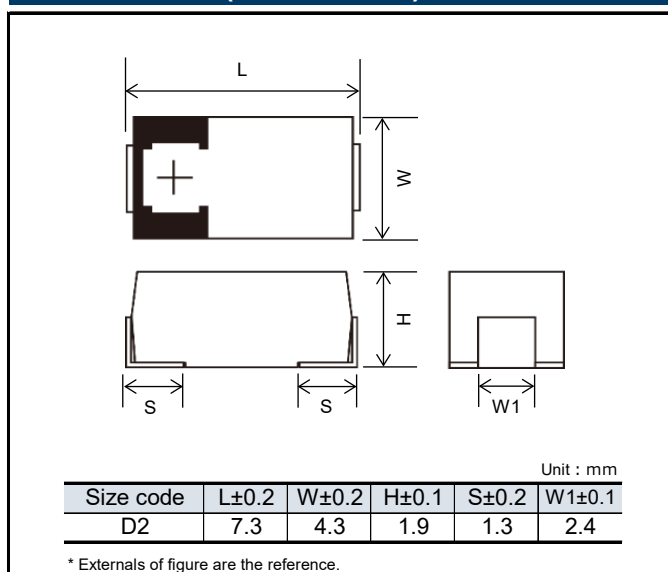
Specifications

Size code	D2	
Category temp. range	-55 °C to +105 °C	
Rated voltage range	10 V	
Category voltage range	10 V	
Rated capacitance range	68 µF to 100 µF	
Capacitance tolerance	±20 % (120 Hz / +20 °C)	
Leakage current	Please see the attached characteristics list	
Dissipation factor(tan δ)	Please see the attached characteristics list	
Surge voltage (V)	Rated voltage × 1.15	
Endurance	+105 °C 2000 h , rated voltage applied	
	Capacitance change	Within ±20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	Within the initial limit
Damp heat (Steady State)	+60 °C, 90 % to 95 % RH, 500 h, No-applied voltage	
	Capacitance change	Within +40 %, -20 % of the initial value
	Dissipation factor(tan δ)	≤ 1.5 times of the initial limit
	Leakage current	≤ 3 times of the initial limit

Marking



Dimensions (not to scale)



Characteristics list

Rated voltage (V)	Rated temperature (°C)	Category voltage (V)	Category temperature (°C)	Rated capacitance (μF)	Case size (mm)			Size code	Specifications				Standard		Floor life level	
					L	W	H		Ripple current (mA rms)	ESR* ² (mΩ max.)	tan δ* ³	LC* ⁴ (μA)	Part number	Min. packaging q'ty (pcs)	Reflow temp ≤260°C	Reflow temp ≤250°C
10	NRFND 105	10	105	68	7.3	4.3	1.9	D2	1700	45	0.10	68.0	10TPC68M	3000	3	2a
	105	10	105		7.3	4.3	1.9		1450	60	0.10	68.0	10TPC68MM	3000		
	105	10	105		7.3	4.3	1.9		1100	100	0.10	68.0	10TPC68ME	3000		
	105	10	105	100	7.3	4.3	1.9		1700	45	0.10	100.0	10TPC100M	3000		
	105	10	105		7.3	4.3	1.9		1600	50	0.10	100.0	10TPC100MT	3000		
	105	10	105		7.3	4.3	1.9		1500	55	0.10	100.0	10TPC100MG	3000		
	105	10	105		7.3	4.3	1.9									

*1: Ripple current (100 kHz / +45 °C)

*2: ESR (100 kHz / +20 °C)

*3: tan δ (120 Hz / +20 °C)

*4: After 5 minutes

NRFND Not recommended for new design

◆ Please refer to each page in this catalog for "Reflow conditions", "Taping specifications" and "Floor life level".

EOL Models (1)

The following table is a list of the End-Of-Life (EOL) models.

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Series	Size code	Models for deletion	Year of deletion	Alternative model	Series	Size code	Models for deletion	Year of deletion	Alternative model
TPSF	B2S	2TPSF270MC	2012/9	2TPSF270M9G	TQC	B15	35TQC2R7MYF	2016	—
		2TPSF270M9	2012/9	2TPSF270M9G			16TQC10M	2023	35TQC10MYFB
	B1S	ETPSF200M9ED	2014	—		B2	20TQC8R2M	2023	35TQC10MYFB
TPE	B2	2R5TPE220MPB	2012/9	2R5TPE220MLB			25TQC5R6M	2023	35TQC10MYFB
		2R5TPE220MDGB	2013	2R5TPE220MFGB			35TQC3R9MYF	2023	35TQC10MYFB
		2TPE330MFB	2021	ETPE330MFB		D12	16TQC33MYFS	2024	20TQC33MD2
		2TPE330MAFB	2021	ETPE330MFB			16TQC47MW	2023	16TQC47MD2
		2TPE330MADGB	2021	ETPE330M9GB			25TQC15MV	2023	25TQC15MD2
		ETPE330MAFB	2021	ETPE330MFB		D2	25TQC22MV	2023	25TQC22MD2
							35TQC10M	2024	35TQC10MD2
	C2	2R5TPE330MFC2	2012/9	2R5TPE330MF	TA	D2E	10TAE68M	2023	—
		2R5TPE330MIC2	2012/9	2R5TPE330MF			4TPB100MV	2012/9	4TPE100MZB
		2R5TPE330MCC2	2012/9	2R5TPE330MC	TPB	B2	4TPB150MA	2012/9	4TPE150MAZB
		2R5TPE330M9C2	2012/9	2R5TPE330M9			8TPB33M	2012/9	10TPB33M
		4TPE220MFC2	2012/9	4TPE220MF			2R5TPB220MC	2012/9	4TPE220MI
		4TPE220MIC2	2012/9	4TPE220MI		C	4TPB150MC	2012/9	6TPE150M
		4TPE220MPC2	2012/9	4TPE220MI			4TPB220MC	2012/9	4TPE220MI
		6TPE150MIC2	2012/9	6TPE150MI			6TPB100MC	2012/9	6TPS150MUD
		6TPE150MPC2	2012/9	6TPE150M			6TPB150MC	2012/9	6TPE150M
		8TPE100MPC2	2012/9	10TPF150ML			8TPB82MC	2012/9	8TPE100MAZB
	C3	2R5TPE330MFC	2012/9	2R5TPE330MF			10TPB47MC	2012/9	10TPC100M
		2R5TPE330MIC	2012/9	2R5TPE330MF			10TPB68MC	2012/9	10TPC100M
		2R5TPE330MPC	2012/9	2R5TPE330MF			10TPB220MC	2009/10	—
		4TPE220MIC	2012/9	4TPE220MI		D3L	2R5TPB470ML	2012/9	2R5TPE470MI
		4TPE220MPC	2012/9	4TPE220MI			2R5TPB680ML	2012/9	2R5TPE680MFL
		6TPE150MPC	2012/9	6TPE150M			16TPB47ML	2003/6	16TQC47MD2
		6TPE220MIC	2012/9	6TPE220MI		D3	2R5TPB330M	2012/9	2R5TPE330MI
		6TPE220MPC	2012/9	6TPE220M			16TPB47M	2003/6	16TQC47MD2
		10TPE150MGC	2012/9	10TPE220ML	TC	D3L	6TCF220M5L	2025	—
		10TPE180MGC	2012/9	10TPE220ML		D4	ETCF680M5H	2026	ETCF680M6L
	D15E	6TPE470MAZU	2024	6TPE470M	TPC	C1	2R5TPC82M	2012/9	—
	D2E	2TPE330M6	2011	ETPSF270M6E			4TPC56M	2012/9	—
		2TPE330M7	2011	2R5TPE330M7			4TPC100M	2012/9	6TPS150MUD
		2TPE330M9	2011	2R5TPE330M9			6TPC68M	2012/9	6TPS150MUD
		2TPE470M6	2011	2R5TPF470M6L			6TPC100MC	2012/9	6TPS150MUD
		2TPE470M7	2011	2R5TPE470M7			8TPC33M	2012/9	16TQC33MB2
		2TPE470M9	2011	2R5TPE470M9		B1	6TPC47M	2024	10TPE47MAZB
		2R5TPE220M	2026	2R5TPE220MI			6TPC47MB	2024	10TPE47MAZB
		2R5TPE330M	2026	2R5TPE330MI			8TPC22M	2024	20TQC22MB2
	D3L	6TPE330MAL	2021	6TPE330ML			8TPC150M	2024	—
		6TPE330MA9EL	2026	6TPF330M9L			12TPC10M	2024	35TQC10MYFB
TPF	D2E	2TPF470M6	2012/9	2R5TPF470M6L		D2	12TPC15M	2024	25TQC15MB
		2TPF220M6	2024	ETPSF270M6E			2R5TPC220M	2012/9	2R5TPE220MI
		2TPF330M6	2024	ETPSF270M6E			16TPC33M	2003/6	20TQC33MD2
	D3L	2R5TPF330M7L	2023	2R5TPF470M7L			6TPC100M	2024	6TPE100MW
		4TPF470M5EL	2014	—			6TPC150M	2024	6TPE150M
		6TPF330M5EL	2014	—			6TPC330MA	2024	6TPE330MAP
		6TPF220M5L	2025	6TPF330M9L					
TQS	B1S	16TQS33MBD	2020	—					
		25TQS10MED	2020	—					
		35TQS6R8MHD	2020	—					

EOL Models (2)

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Series	Size code	Models for deletion	Year of deletion	Alternative model	Series	Size code	Models for deletion	Year of deletion	Alternative model	
APA	D2A	APA series	2006/4	—	TPLF	D2T	2TPLF470M7	2012/9	—	
APB	D1	APB series	2006/4	—			2TPLF560M6	2011/7	—	
APC	D2	APC series	2009/6	—			All models	2013	—	
APD	D1	APD series	2009/6	—	TPU	S08	2R5TPU22M	2012/9	—	
TPA	C	6TPA47M	2012/9	10TPB47M			4TPU15M	2012/9	—	
		10TPA33M	2012/9	10TPB33M			6TPU10M	2012/9	—	
	D3	4TPA220M	2012/9	4TPE220M		2R5TPU22MSI	2011	—		
		6TPA150M	2012/9	6TPE150M		2R5TPU47MSI	2018	—		
		10TPA100M	2012/9	10TPC100M		ETPU100MSI	2018	—		
TPD	D4D	2R5TPD470M	2007/10	2R5TPF470ML		S09	4TPU15MSI	2011	—	
		2R5TPD470M5	2012/3	2R5TPE470M7			4TPU33MSI	2011	—	
		2R5TPD470M6	2012/3	2R5TPF470M6L			4TPU68MSI	2018	—	
		2R5TPD470M8	2007/10	2R5TPF470M7L			6TPU10MSI	2018	—	
		2R5TPD680M	2007/10	2R5TPF680ML			6TPU22MSI	2018	—	
		2R5TPD680M5	2012/3	2R5TPF680M6L			6TPU47MSI	2018	—	
		2R5TPD680M6	2012/3	2R5TPF680M6L			10TPU4R7MSI	2018	—	
		2R5TPD680M8	2007/10	2R5TPF680M7L			S11	2R5TPU47MSK	2012/9	—
		2R5TPD1000M	2012/3	ETPF1000M6H				2R5TPU68MSK	2013	—
		2R5TPD1000M8	2012/3	ETPF1000M6H				4TPU33MSK	2012/9	—
		2R5TPD1000M6	2012/3	ETPF1000M6H		4TPU47MSK		2013	—	
		2R5TPD1000M5	2012/3	ETPF1000M5H		6TPU22MSK		2012/9	—	
		4TPD330M	2007/10	4TPF330ML		6TPU33MSK		2013	—	
		4TPD470M	2007/10	4TPF470ML		A09	2R5TPU100MAI	2011	—	
		4TPD680M	2012/3	4TPF680MAH			4TPU68MAI	2011	—	
		6TPD220M	2007/10	6TPF220ML			6TPU47MAI	2011	—	
		6TPD330M	2007/10	6TPF330M9L			10TPU33MAI	2011	—	
		6TPD470M	2012/3	6TPF470MAH		B09	4TPU68MBI	2012/9	—	
		10TPD150M	2007/10	10TPF150ML			6TPU47MBI	2012/9	—	
			6TPU150MBI	2018	—					
TPG	B1G	4TPG150M	2012/9	6TPS150MUD	TH	D2	8TPU33MBI	2012/9	—	
		6TPG68MG	2012/9	6TPS150MUD			2R5THC220M	2026	4TCE220M	
		2R5TPG220M	2021	ETPS220MUD			6THC150M	2024	6TCE150M	
		6TPG100M	2021	6TPS150MUD		10THC68M	2026	—		
		6TPG100MG	2021	6TPS150MUD		D2E	2R5THE330MF	2026	4TCE220MF	
		6TPG100MZGD	2021	6TPS150MUD			2R5THE330MI	2026	4TCE220MI	
		8TPG47M	2021	10TPE47MAZB			2R5THE330M	2026	4TCE220M	
		10TPG33M	2011	16TQC33MB2			4THE220MF	2026	4TCE220MF	
		10TPG47M	2024	10TPE47MAZB			4THE220MI	2026	4TCE220MF	
		12TPG33M	2024	16TQC33MB2			4THE220M	2026	4TCE220MF	
		B15G	6TPG220MZG	2014			—	6THE150MI	2026	6TCE150MF
			2R5TPG220MUG	2021			ETPS220MUD	6THE150M	2026	6TCE150MF
			4TPG220M	2021		4TPS220MUD	D3L	2R5THB470ML	2012/9	6TCE470M
			6TPG150M	2021		6TPS150MUD		4THB330ML	2026	4TCE330M
			6TPG150MZG	2021		6TPS150MUD		6THB220ML	2026	6TCE220M
TPH	A09	ETPH100MHA	2018	—		D4	2R5THB680M	2012/9	—	
		4TPH68MHA	2018	—			2R5THB1000M	2012/9	—	
		6TPH47MHA	2018	—			6THB330M	2026	6TCE330ML	
		6TPH100MAEA	2018	—			6THB470M	2026	6TCE470M	
		ATPH33MAHA	2018	—			10THB220M	2026	10TCE220ML	
	A14	ETPH220MABC	2018	—	10THB330M		2026	10TCE330M		
		ETPH220MAZC	2013	—	D4D	2R5THD680M	2012/3	2R5TPF680M6L		
		4TPH150MABC	2018	—		4THD470M	2012/9	—		
		6TPH100MABC	2018	—		6THD330M	2012/3	6TPF330M9L		
TLE	B2	ETLE330MCGB	2024/4	ETPE330M9GB						
		6TLE100MZB	2024/4	6TPE100MZB						
TR		TR series	—	—						
TPL	D2T	2R5TPL220MC	2012/9	—						
		2R5TPL330M7	2011/7	—						
		All models	2013	—						
	D12T	All models	2013	—						
	D15T	All models	2013	—						

Safety Precautions

When using our products, no matter what sort of equipment they might be used for,
be sure to confirm the applications and environmental conditions with our specifications in advance.

Panasonic
INDUSTRY

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